INCH-POUND

MIL-P-29590/1
29 November 1991

# MILITARY SPECIFICATION SHEET

## POWER SUPPLIES, AIRBORNE, ELECTRONIC

## 5.0 VOLT, 40 AMPERE, KEY CODE A88UG

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this associated detail specification and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-P-29590.

1. SCOPE

1.1 Scope. This document, along with MIL-P-29590, establishes the requirements and test procedures for an electronic power supply with a single 5 volt output.

1.2 Classification. This specification covers family A2 power supplies in accordance with MIL-P-29250.

AMSC N/A FSC 6130
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## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

2.1.1 Specification, standards, and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION
MILITARY

| | |
|---|---|
| MIL-P-29590 - | Power Supplies, Airborne, Electronic, General Specification for. |

STANDARDS
MILITARY

| | |
|---|---|
| MIL-STD-202 - | Test Methods for Electronic and Electrical Component Parts. |
| MIL-STD-461 - | Electromagnetic Emission and Susceptibility Requirements for the control of Electromagnetic Interference. |
| MIL-STD-462 - | Electromagnetic Interference Characteristics, Measurement of. |
| MIL-STD-704 - | Aircraft Electric Power Characteristics. |
| MIL-STD-1389 - | Design Requirements for Standard Electronic Modules. |
| MIL-STD-1665 - | Test Equipment for the Standard Electronic Modules Program. |

HANDBOOK
MILITARY

| | |
|---|---|
| MIL-HDBK-241 - | Design Guide for Electromagnetic Interference (EMI) Reduction in Power Supplies. |



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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. Power supplies shall comply with the family A2 requirements of MIL-P-29590 except as specified herein.

3.2 Correlation requirements. Before any production power supplies are tested, all testers and test equipment shall be correlated in accordance with MIL-P-29590.

3.3 Reliability requirements. Power supplies shall have a minimum mean time between failures (MTBF) of 18,000 hours for airborne uninhabited fighter environments when predicted in accordance with MIL-P-29590. The prediction shall be made with a rib temperature of 85 degrees Celsius (°C).

3.4 Electrical performance requirements. Power supplies shall conform to the electrical requirements specified in table I and shall operate within the environments specified in MIL-P-29590.



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TABLE I. Power supply electrical requirements.

| Test cond | Ref test table | Input voltage DC | Parameters | Test type (see notes 1/-3/) | Acceptance requirements 25 ± 5°C rib temperature | | | | -55°C to 85°C rib temperature | | | | Units 5/ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Initial | | End of life | | Initial | | End of life | | |
| | | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| A1, A2 | III | - | Isolation-input to chassis 600 VDC | X | 10.0 | INF | 1.0 | INF | 10.0 | INF | 1.0 | INF | MOhm |
| A3-A6 | III | - | Isolation-input to output 600 VDC | X | 10.0 | INF | 1.0 | INF | 10.0 | INF | 1.0 | INF | MOhm |
| B1 | IV | - | Chassis pin to chassis continuity | X | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | Ohm |
| B2-B54 | IV | - | Parallel connector pin continuity | X | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | Ohm |
| C1, C2 | V | 270 | Load regulation, 5V, 4A to 40A | X | 4.950 | 5.050 | 4.925 | 5.075 | 4.900 | 5.100 | 4.875 | 5.125 | VDC |
| C3, C4 | V | 270 | Load regulation, 5V, no load to 4A | X | 4.900 | 5.100 | 4.875 | 5.125 | 4.850 | 5.150 | 4.825 | 5.175 | VDC |
| D1 | VI | 240 | Line regulation, 4A | X | 4.950 | 5.050 | 4.925 | 5.075 | 4.900 | 5.100 | 4.875 | 5.125 | VDC |
| D2 | VI | 240 | Line regulation, 40A | X | 4.950 | 5.050 | 4.925 | 5.075 | 4.900 | 5.100 | 4.875 | 5.125 | VDC |





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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Initial | | End of life | | Initial | | End of life | | |
| | | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| D3 | VI | 280 | Line regulation, 4A | X | 4.950 | 5.050 | 4.925 | 5.075 | 4.900 | 5.100 | 4.875 | 5.125 | VDC |
| D4 | VI | 280 | Line regulation, 40A | X | 4.950 | 5.050 | 4.925 | 5.075 | 4.900 | 5.100 | 4.875 | 5.125 | VDC |
| E1 | VII | 240 | Low input voltage protection | Q | - | 1.8 | - | 1.8 | - | 1.8 | - | 1.8 | - |
| G1 | VIII | 270 | Efficiency, 40A | X | 80 | - | 80 | - | 80 | - | 80 | - | % |
| G2 | VIII | 270 | Efficiency, 20A | Q | 80 | - | 80 | - | 80 | - | 80 | - | % |
| K1 | IX | 270 | Inrush current | Q | - | 100 | - | 100 | - | 100 | - | 100 | Apk |
| L1 | X | 270 | Remote sense compensation | Q | 4.950 | 5.050 | 4.925 | 5.075 | 4.900 | 5.100 | 4.875 | 5.125 | VDC |
| N1 | XI | 270 | Overload protection | X | 4.950 | 5.050 | 4.925 | 5.075 | 4.900 | 5.100 | 4.875 | 5.125 | VDC |
| N2 | XI | 270 | Overload protection | X | 44 | 50 | 44 | 50 | 44 | 50 | 44 | 50 | ADC |





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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Initial | | End of life | | Initial | | End of life | | |
| | | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| N3 | XII | 270 | Short circuit recovery time | Q | 0 | 100 | 0 | 100 | 0 | 100 | 0 | 100 | msec |
| N4 | XIII | 270 | Short circuit overshoot | Q | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | V |
| N5 | XIV | 270 | Short circuit current | S | 0 | 50 | 0 | 50.8 | 0 | 50 | 0 | 50.8 | ADC |
| P1 | XV | 270 | Turn-on transient | Q | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | V |
| P2 | XV | 270 | Turn-on time | S | 0 | 250 | 0 | 250 | 0 | 250 | 0 | 250 | msec |
| P3 | XVI | 270 | Turn-off transient | Q | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | V |
| P4 | XVII | 270 | Control turn-on transient | Q | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | V |
| P5 | XVII | 270 | Control turn-on time | S | 0 | 250 | 0 | 250 | 0 | 250 | 0 | 250 | msec |
| P6 | XVII | 270 | Control turn-off | Q | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | V |





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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Initial | | End of life | | Initial | | End of life | | |
| | | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| Q1, Q2 | XVIII | 240, 280 | Ripple, total, mVp-p (10Hz to 5MHz) 4A | X | 0 | 130 | 0 | 150 | 0 | 130 | 0 | 150 | mVp-p |
| Q3, Q4 | XVIII | 240, 280 | Ripple, total, mVp-p (10Hz to 5MHz) 40A | S | 0 | 130 | 0 | 150 | 0 | 130 | 0 | 150 | mVp-p |
| Q5 | XIX | 270 | Ripple, total over-current (10Hz to 5MHz) > 40A | Q | 0 | 200 | 0 | 200 | 0 | 200 | 0 | 200 | mVp-p |
| Q6 | XIX | 270 | Ripple, total, under-current (10Hz to 5MHz) < 4A | Q | 0 | 200 | 0 | 200 | 0 | 200 | 0 | 200 | mVp-p |
| Q7, Q8 | XX | 240, 280 | Ripple, total (10Hz to 20MHz) 40A | Q | 0 | 300 | 0 | 300 | 0 | 300 | 0 | 300 | mVp-p |
| R1 | XXI | 270 | Dynamic line regulation, +21% | Q | 0 | 150 | 0 | 150 | 0 | 150 | 0 | 150 | mV |





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| Test cond | Ref test table | Input voltage DC | Parameters | Test type (see notes 1/-3/) | Acceptance requirements 25 ± 5°C rib temperature | | | | -55°C to 85°C rib temperature | | | | Units 5/ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Initial | | End of life | | Initial | | End of life | | |
| | | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| R2 | XXI | 270 | Dynamic line ringing, +21% (2nd half cycle) | Q | 0 | 75 | 0 | 75 | 0 | 75 | 0 | 75 | mV |
| R3 | XXII | 213.3 | Dynamic line regulation, -21% | Q | 0 | 150 | 0 | 150 | 0 | 150 | 0 | 150 | mV |
| R4 | XXII | 213.3 | Dynamic line ringing, -21% (2nd half cycle) | Q | 0 | 75 | 0 | 75 | 0 | 75 | 0 | 75 | mV |
| S1 | XXIII | 270 | Dynamic load regulation, 80 percent to 90 percent | X | 0 | 150 | 0 | 150 | 0 | 150 | 0 | 150 | mV |
| S2 | XXIII | 270 | Dynamic load ringing, (2nd half cycle) | S | 0 | 75 | 0 | 75 | 0 | 75 | 0 | 75 | mV |





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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Initial | | End of life | | Initial | | End of life | | |
| | | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| S3 | XXIII | 270 | Dynamic load regulation, 90 percent to 80 percent | X | 0 | 150 | 0 | 150 | 0 | 150 | 0 | 150 | mV |
| S4 | XXIII | 270 | Dynamic load ringing, (2nd half cycle) | S | 0 | 75 | 0 | 75 | 0 | 75 | 0 | 75 | mV |
| S5 | XXIII | 270 | Dynamic load regulation, 70 percent to 90 percent | Q | 0 | 150 | 0 | 150 | 0 | 150 | 0 | 150 | mV |
| S6 | XXIII | 270 | Dynamic load ringing, (2nd half cycle) | Q | 0 | 75 | 0 | 75 | 0 | 75 | 0 | 75 | mV |
| S7 | XXIII | 270 | Dynamic load regulation, 90 percent to 70 percent | Q | 0 | 150 | 0 | 150 | 0 | 150 | 0 | 150 | mV |





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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Initial | | End of life | | Initial | | End of life | | |
| | | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| S8 | XXIII | 270 | Dynamic load ringing, (2nd half cycle) | Q | 0 | 75 | 0 | 75 | 0 | 75 | 0 | 75 | mV |
| S9 | XXIV | 270 | Dynamic load regulation, 30 percent to 80 percent full capacitor load | Q | 0 | 150 | 0 | 150 | 0 | 150 | 0 | 150 | mV |
| S10 | XXIV | 270 | Dynamic load ringing, (2nd half cycle) | Q | 0 | 75 | 0 | 75 | 0 | 75 | 0 | 75 | mV |
| T | XXV | 270 | Turn-on time into capacitive load | Q | 0 | 250 | 0 | 250 | 0 | 250 | 0 | 250 | msec |
| V1 | XXVI | 270 | Output status signal, logic low, 1A | X | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | VDC |





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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Initial | | End of life | | Initial | | End of life | | |
| | | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| V2 | XXVI | 270 | Output status signal, logic low, 40A | X | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | VDC |
| V3 | XXVII | 270 | Undervoltage - output status signal, logic high, 1A | X | 29.0 | 30.5 | 29.0 | 30.5 | 29.0 | 30.5 | 29.0 | 30.5 | VDC |
| V4 | XXVII | 270 | Overvoltage - output status signal logic high, 40A | X 4/ | 29.0 | 30.5 | 29.0 | 30.5 | 29.0 | 30.5 | 29.0 | 30.5 | VDC |
| X | XXVIII | 175 | Output voltage hold-up time | Q | 50 | - | 50 | - | 50 | - | 50 | - | µsec |
| Y | XXIX | - | Input to chassis capacitance | S | - | 0.004 | - | 0.004 | - | 0.004 | - | 0.004 | µF |
| Z | XXX | 270 | Output impedance, 10Hz to 25KHz, 100 percent rated load | Q | 0 | 30 | 0 | 30 | 0 | 30 | 0 | 30 | mOhm |





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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Initial | | End of life | | Initial | | End of life | | |
| | | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| BB1 | XXXI | 280 | Line transient response (high line) initial transient | Q | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | V |
| BB2 | XXXI | 280 | Line transient response (high line) 2nd half cycle | Q | -0.13 | 0.13 | -0.13 | 0.13 | -0.13 | 0.13 | -0.13 | 0.13 | V |
| BB3 | XXXI | 240 | Line transient response (low line) initial transient | Q | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | -0.25 | 0.25 | V |
| BB4 | XXXI | 240 | Line transient response (low line) 2nd half cycle | Q | -0.13 | 0.13 | -0.13 | 0.13 | -0.13 | 0.13 | -0.13 | 0.13 | V |
| CC1 | XXXII | 270 | Remote ON/OFF signal logic low | Q | 4.950 | 5.050 | 4.925 | 5.075 | 4.900 | 5.100 | 4.875 | 5.125 | VDC |
| CC2 | XXXII | 270 | Remote ON/OFF signal logic high | Q | 0 | 0.4 | 0 | 0.4 | 0 | 0.4 | 0 | 0.4 | VDC |





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| Test cond | Ref test table | Input voltage DC | Parameters | Test type (see notes 1/-3/) | Acceptance requirements 25 ± 5°C rib temperature | | | | -55°C to 85°C rib temperature | | | | Units 5/ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Initial | | End of life | | Initial | | End of life | | |
| | | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| CC3 | XXXII | 270 | Remote ON/OFF signal shorted | Q | 4.950 | 5.050 | 4.925 | 5.075 | 4.900 | 5.100 | 4.875 | 5.125 | VDC |
| CC4 | XXXIII | 270 | Over-temperature shutdown, logic high | Q | 4.950 | 5.050 | 4.925 | 5.075 | 4.900 | 5.100 | 4.875 | 5.125 | VDC |
| CC5 | XXXIII | 270 | Over-temperature shutdown, logic low | Q | 0 | 0.4 | 0 | 0.4 | 0 | 0.4 | 0 | 0.4 | VDC |
| EE1 | XXXIV | 270 | Common mode current, broadband 40A | Q | 0 | 200 | 0 | 200 | 0 | 200 | 0 | 200 | mAp-p |
| EE2 | XXXIV | 270 | Common mode current, broadband 20A | Q | 0 | 200 | 0 | 200 | 0 | 200 | 0 | 200 | mAp-p |
| GG1 | XXXV | 270 | Input voltage turn-on threshold | S | 200 | 220 | 200 | 220 | 200 | 220 | 200 | 220 | VDC |
| GG2 | XXXV | 270 | Input voltage turn-off threshold | S | - | 165 | - | 165 | - | 165 | - | 165 | VDC |





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NOTES:

1/ "X" parameters shall be 100 percent tested during power supply quality conformance testing.

2/ "S" parameters shall be sample tested (one percent or two power supplies from each inspection lot, whichever is greater) during quality conformance testing.

3/ "Q" parameters shall be tested during qualification testing.

4/ Proper operation of the overvoltage protection circuitry shall be verified as an "X" test by the power supply vendor during manufacture of the power supplies.

5/ The symbols used in this table and not defined elsewhere are as follows:

A - amperes
ADC - amperes direct current
Apk - amperes peak
GND - ground
IIN - input current
IOUT - output current
mAp-p - milliamperes peak-to-peak
mOhm - milliohm
msec - millisecond
INF - infinite
MOhm - megohm
mV - millivolt
mVpk - millivolt peak
mVp-p - millivolt peak-to-peak
% - percent
RET - return
TON - turn-on time
tR1 - rise time
µADC - microamperes direct current
V - volts
VDC - volts direct current
VOUT - output voltage
Zi - input impedance



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3.4.1 Input power. The power supply shall meet the input power requirements of MIL-P-29590 except as modified herein. The power supply input power shall be 270 volts direct current (VDC) nominal derived from full wave rectified and filtered, 3 phase alternating current (AC) power, where the AC power has characteristics as specified in MIL-STD-704. The ripple voltage between the frequencies of 2160 hertz (Hz) and 2640 Hz shall be 20 Vp-p maximum.

3.4.1.1 Normal operation. Normal operation shall apply when the input voltage has the characteristics specified herein. The steady state voltage shall be 240 VDC minimum and 280 VDC maximum. The normal transient voltage shall be within the envelope on figure 1.

3.4.1.2 Emergency operation. Not applicable.

3.4.1.3 Transient operation. The transient characteristics shall be as shown on figure 1.

3.4.1.4 Abnormal operation. The power supply shall be permitted degraded performance or loss of function when the input power has voltage characteristics below the voltage transient levels shown on figure 1. The power supply shall not produce a damaging or unsafe condition during abnormal operation. The power supply shall automatically recover full specified performance within 250 msec after the input power characteristics are restored to the normal operation limits.

3.4.1.5 Auxiliary DC input. Not applicable.

3.4.1.6 Turn-on threshold. The turn-on threshold shall be between 200 VDC and 220 VDC.

3.4.1.7 Turn-on to turn-off hysteresis. The turn-off threshold shall be 165 VDC.

3.4.1.8 Step input. Upon application of a 270 VDC step input, the inrush current shall not exceed 100A. The product of current squared and time shall not exceed 0.2A squared-seconds from the time the step input voltage is applied until the time that the input current is within 200 percent of its full load, steady state value. Subsequent to the initial current surge, or upon subsequent enabling of the power converter, the input current shall not exceed 125 percent of its full load, steady state value. The rate of increase in input current during the subsequent surge shall not exceed 1A per millisecond.



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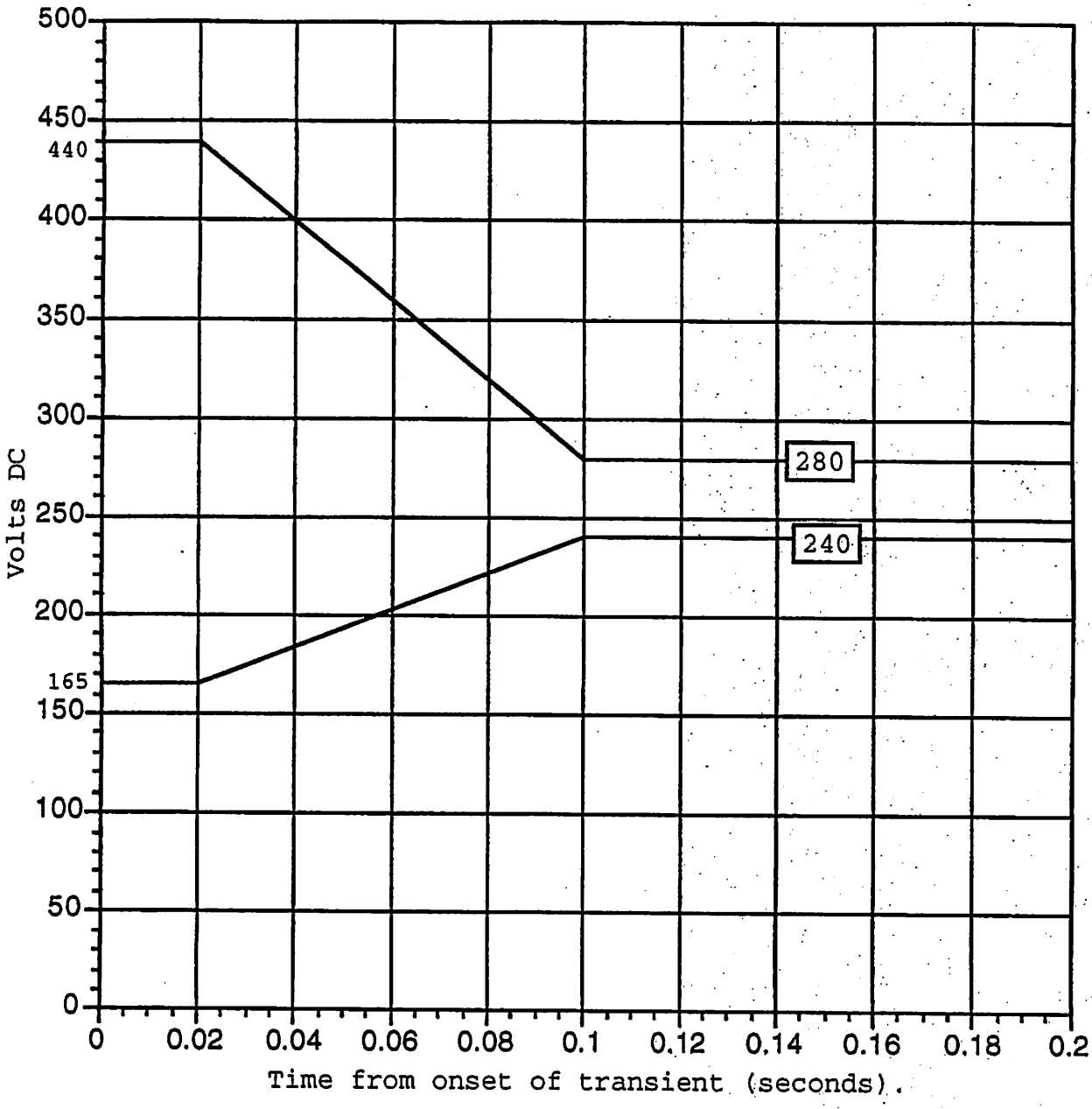



FIGURE 1. Normal operating transient voltage envelope.



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3.4.1.9 Harmonic current limitation. Not applicable.

3.4.1.10 Power factor. Not applicable.

3.4.2 Output power. The output power shall be nominal 5 volts and 40 amperes.

3.4.3 Remote voltage sensing. The power supply shall have one pair of remote voltage sensing leads capable of compensating for a total voltage drop of 200 millivolts in the power output and return. If the remote sense leads are not connected, the output voltage shall be sensed internal to the power supply.

3.4.4 Output impedance. The output impedance shall not exceed 30 milliohms over the frequency range of 10 Hz to 25 KHz.

3.4.5 Parallel operation. The power supply shall meet all performance requirements when operating in parallel with up to four other power supplies under the conditions specified herein. Failure modes that may cause a system of parallel power supplies to fail shall have a failure rate not greater than 5 percent of the failure rate determined by the mean time between failures (MTBF) of a single power supply (see 3.3). Failure of any power supply (other than a short in the power supply output circuitry) shall not cause failure of the output power bus.

3.4.5.1 Paralleling conditions. Power supplies operating in parallel shall share current within 5A (the difference between the minimum and maximum output currents shall not exceed 5A). The power supplies operating in parallel shall exchange load sharing data by means of a current sharing bus to provide forced current sharing. Current sharing may be degraded if the interface signal fails, however, voltage regulation shall be maintained. Any power supply that fails or is commanded "off" shall not cause the steady state current sharing balance of the remaining paralleled power supplies to exceed the 5A current sharing tolerance. Forced current sharing shall not cause instability of the paralleled power supply configuration.

3.4.5.2 Current share signal. The power supply shall provide a differential voltage signal that is proportional to its output current. This voltage signal shall be 50 millivolts per ampere open circuit. Zero volts shall correspond to an output of zero ampere, and the accuracy shall be within plus or minus 2 amperes from no load to full load. The 3 dB bandwidth of this signal shall be less than 100 Hz. The current share bus driver shall source up to 10 milliamperes (mA) and sink 0 mA current. The output impedance shall be 10 ohms maximum. The current share



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bus receiver shall have an input impedance of 10,000 ohms plus or minus 10 percent. The common mode range of this signal shall be plus or minus 1V with respect to the 5V return. An increase in current share bus voltage shall cause the power supply to increase output current.

3.4.5.3 Parallel load characteristics. The power supply shall operate and maintain static regulation as specified herein when connected to a load capacitance ranging from zero to 12,000 microfarads (uf).

3.4.6 Control signals. Control signals shall be as specified in MIL-P-29590 except as modified herein.

3.4.6.1 Over-temperature shutdown. The power supply shall provide an over-temperature shutdown capability. A logic low shall shut the power supply output off. This input should be left open to disable the over-temperature shutdown.

3.4.7 Status signals. Status signals shall be as specified in MIL-P-29590 except as modified herein.

3.4.7.1 Status signal returns. All status signal returns shall be internally connected at one point.

3.4.7.2 Over-temperature indication. The power supply shall provide a low level signal (0.4 VDC maximum, 5.0 mA minimum) when the temperature measured at the sensor exceeds a temperature proportional to a semiconductor junction temperature of 135°C. A logic high or a high impedance off state shall indicate a semiconductor junction temperature of less than 135°C. The over-temperature output shall be capable of providing automatic shutdown by connecting the complement output pin to the temperature shutdown input pin. The over-temperature indication output shall be capable of driving both the temperature shutdown and the external indication circuitry.

3.4.7.3 Malfunction signal. The power supply shall provide a malfunction signal in accordance with MIL-P-29590.

3.4.7.4 Power interrupt signal. The power supply shall provide a power interrupt signal in accordance with MIL-P-29590 except as modified herein. As the input voltage is applied to the power supply, the power interrupt signal shall go to a low level when the input voltage is greater than 190 VDC but less than or equal to the turn-on threshold specified in 3.4.1.6. The power interrupt signal shall remain low until the input voltage drops to



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between 170 VDC and 175 VDC. A 50 microsecond delay before indicating loss of power is not required.

3.4.8 Efficiency. The power supply efficiency shall be 80 percent minimum when operating from half-load to full load.

3.4.9 Internal frequencies. The power supply internal frequencies shall be 75 KHz or greater.

3.5 Physical requirements.

3.5.1 Size and weight. The power supply configuration requirements shall be in accordance with appendix E of MIL-STD-1389, for module thickness D, except as modified herein. The power supply weight shall be 1.50 pounds maximum.

3.5.2 Lateral displacement. Not required.

3.5.3 Forced air across components cooling. Not required.

3.5.4 Connector location and contact assignment. The power supply connector and connector location shall be in accordance with MIL-STD-1389, appendix E, except as modified herein. The contact quantity shall be five rows of 50 contacts except not all rows shall be complete. Contacts number 41 through 50, 91, 92, 95, 96, 99, 100, 141 through 150, 191, 192, 195, 196, 199, 200, and 241 through 250 shall be removed from the connector body and the resulting holes shall be filled with the specified insulation material.

3.5.5 Module contacts. The number of contacts shall be 208.

3.5.6 Contact assignment. The contact assignment shall be in accordance with table II.



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TABLE II. Connector contact assignment.

| Row A | | Row B | | Row C | | Row D | | Row E | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Output (+5VDC) | 51 | Output (+5VDC) | 101 | Unused | 151 | Unused | 201 | Unused |
| 2 | Output (+5VDC) | 52 | Output (+5VDC) | 102 | Output (+5VDC) | 152 | Unused | 202 | Unused |
| 3 | Output (+5VDC) | 53 | Output (+5VDC) | 103 | Unused | 153 | Unused | 203 | Unused |
| 4 | Output (+5VDC) | 54 | Output (+5VDC) | 104 | Output (+5VDC) | 154 | Unused | 204 | Unused |
| 5 | Output (+5VDC) | 55 | Output (+5VDC) | 105 | Unused | 155 | Unused | 205 | Unused |
| 6 | Output (+5VDC) | 56 | Output (+5VDC) | 106 | Output (+5VDC) | 156 | Unused | 206 | Unused |
| 7 | Output (+5VDC) | 57 | Output (+5VDC) | 107 | Unused | 157 | Unused | 207 | Unused |
| 8 | Output (+5VDC) | 58 | Output (+5VDC) | 108 | Output (+5VDC) | 158 | Unused | 208 | Unused |
| 9 | Output (+5VDC | 59 | Output (+5VDC) | 109 | Unused | 159 | Unused | 209 | Unused |
| 10 | Output ret | 60 | Output ret | 110 | Output (+5VDC) | 160 | Unused | 210 | Unused |
| 11 | Output ret | 61 | Output ret | 111 | Unused | 161 | Unused | 211 | Unused |
| 12 | Output ret | 62 | Output ret | 112 | Output ret | 162 | Unused | 212 | Unused |
| 13 | Output ret | 63 | Output ret | 113 | Unused | 163 | Unused | 213 | Unused |
| 14 | Output ret | 64 | Output ret | 114 | Output ret | 164 | Unused | 214 | Unused |
| 15 | Chassis ground | 65 | Chassis ground | 115 | Chassis ground | 165 | Chassis ground | 215 | Chassis ground |
| 16 | Output ret | 66 | Output ret | 116 | Output ret | 166 | Unused | 216 | Unused |
| 17 | Output ret | 67 | Output ret | 117 | Unused | 167 | Unused | 217 | Unused |
| 18 | Output ret | 68 | Output ret | 118 | Output ret | 168 | Unused | 218 | Unused |
| 19 | Output ret | 69 | Output ret | 119 | Unused | 169 | Unused | 219 | Unused |
| 20 | Output ret | 70 | Output ret | 120 | Output ret | 170 | Over-temp | 220 | Unused |
| 21 | Output ret | 71 | Output ret | 121 | Unused | 171 | Over-temp ret | 221 | Unused |
| 22 | Output ret | 72 | Output ret | 122 | Output ret | 172 | Unused | 222 | Unused |
| 23 | Output ret | 73 | Output ret | 123 | PFI | 173 | Unused | 223 | Unused |
| 24 | Output ret | 74 | Output ret | 124 | Output ret | 174 | Unused | 224 | PFI ret |
| 25 | Output ret | 75 | Output ret | 125 | Malfunction ret | 175 | Unused | 225 | Malfunction |
| 26 | Output ret | 76 | Output ret | 126 | Output ret | 176 | Unused | 226 | Unused |
| 27 | Output ret | 77 | Output ret | 127 | Unused | 177 | Output status | 227 | Output status ret |
| 28 | Output ret | 78 | Output ret | 128 | Output ret | 178 | Remote sense (-) | 228 | Unused |
| 29 | Output (+5VDC) | 79 | Output (+5VDC) | 129 | Unused | 179 | Remote sense (+) | 229 | Unused |
| 30 | Output (+5VDC) | 80 | Output (+5VDC) | 130 | Output (+5VDC) | 180 | Current share ret | 230 | Current share |
| 31 | Output (+5VDC) | 81 | Output (+5VDC) | 131 | On off | 181 | Unused | 231 | Unused |



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TABLE II Connector contact assignment - Continued.

| Row A | | Row B | | Row C | | Row D | | Row E | |
|---|---|---|---|---|---|---|---|---|---|
| 32 | Output (+5VDC) | 82 | Output (+5VDC) | 132 | Output (+5VDC) | 182 | Unused | 232 | On off ret |
| 33 | Output (+5VDC) | 83 | Output (+5VDC) | 133 | Over-tempera-ture shutdown ret | 183 | Unused | 233 | Over-tempera-ture shutdown |
| 34 | Output (+5VDC) | 84 | Output (+5VDC) | 134 | Output (+5VDC) | 184 | Unused | 234 | Unused |
| 35 | Output (+5VDC) | 85 | Output (+5VDC) | 135 | Unused | 185 | Unused | 235 | Unused |
| 36 | Output (+5VDC) | 86 | Output (+5VDC) | 136 | Output (+5VDC) | 186 | Unused | 236 | Unused |
| 37 | Output (+5VDC) | 87 | Output (+5VDC) | 137 | Unused | 187 | Unused | 237 | Unused |
| 38 | Unused | 88 | Unused | 138 | Unused | 188 | Unused | 238 | Unused |
| 39 | Unused | 89 | Unused | 139 | Unused | 189 | Unused | 239 | Unused |
| 40 | Chassis ground | 90 | Chassis ground | 140 | Chassis ground | 190 | Chassis ground | 240 | Chassis ground |
| 41 | Pin removed | 91 | Pin removed | 141 | Pin removed | 191 | Pin removed | 241 | Pin removed |
| 42 | Pin removed | 92 | Pin removed | 142 | Pin removed | 192 | Pin removed | 242 | Pin removed |
| 43 | Pin removed | 93 | 270 VDC | 143 | Pin removed | 193 | Unused | 243 | Pin removed |
| 44 | Pin removed | 94 | 270 VDC | 144 | Pin removed | 194 | Unused | 244 | Pin removed |
| 45 | Pin removed | 95 | Pin removed | 145 | Pin removed | 195 | Pin removed | 245 | Pin removed |
| 46 | Pin removed | 96 | Pin removed | 146 | Pin removed | 196 | Pin removed | 246 | Pin removed |
| 47 | Pin removed | 97 | 270 VDC ret | 147 | Pin removed | 197 | Unused | 247 | Pin removed |
| 48 | Pin removed | 98 | 270 VDC ret | 148 | Pin removed | 198 | Unused | 248 | Pin removed |
| 49 | Pin removed | 99 | Pin removed | 149 | Pin removed | 199 | Pin removed | 249 | Pin removed |
| 50 | Pin removed | 100 | Pin removed | 150 | Pin removed | 200 | Pin removed | 250 | Pin removed |



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3.6 Environmental requirements. The power supply shall meet the environmental requirements specified in MIL-P-29590 except as modified herein.

3.6.1 Temperature and altitude. The power supply shall meet the temperature and altitude requirements as specified in MIL-P-29590 except the barometric pressure shall be 33 millimeters of mercury. This pressure corresponds to an altitude of 70,000 feet.

3.7 Design of the power supply for stability. The power supply is not required to meet the full requirements of MIL-STD-461. Therefore, for a power system to meet MIL-STD-461 conducted emissions requirement CE03, the system must have an electromagnetic interference (EMI) filter between the power source output and the power supply input. The necessary EMI filter attenuation can be found by taking the difference between the allowed applicable power supply conducted emissions curves and the system required conducted emissions. The power supply input impedance characteristics as specified in 3.7.1 are also needed in order to design an EMI filter. MIL-HDBK-241 should be referenced for precautions to be taken in the design of an EMI filter to prevent the filter/power supply combination from forming a negative impedance oscillator. The design of the power supply must accommodate the worst case minimum input voltage, including low line transient voltage.

3.7.1 Input impedance requirement. The power supply input impedance magnitude, at full load output, shall be greater than the figure 2 curve for the input voltage specified. The input impedance phase shall be more positive than the figure 3 curve for the input voltages specified. These curves are composites of the minimum input impedances from several different power supply designs. The actual model for each vendor's design may be obtained from the Airborne Power Supply Products Quality Assurance Activity (APSP-QAA). During initial qualification testing, the APSP-QAA will verify that each vendor's design meets the above impedance requirements.

3.7.2 Input impedance computer model. Each manufacturer of power supplies shall generate a computer model of the power supply input impedance (Zi) as observed from the input power contacts. The model shall diagram the input network and show both the AC and auxiliary inputs. It shall include elements simulating the negative impedance of the converter. Nominal component values, together with their tolerances, shall be given. To verify the accuracy of the model, each manufacturer shall measure the input



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impedance (magnitude and phase) of their power supply under the following conditions:

(a) Over the frequency range of 100 Hz to 100 KHz or up to the voltage feedback loop gain crossover frequency, whichever is greater.

(b) With input voltage of 165 and 270 VDC. Also measurement at a minimum input voltage, just prior to the output voltage going out of regulation or turning off.

(c) At full load for all input voltages. For 165 and 270 VDC input voltage, also measure at 10 percent and 50 percent full load.

The Zi computer model shall be made to match the Zi measured data within ± 1.0 decibel (dB) for gain and ± 6 degrees for phase at frequencies up to 10 KHz. Above 10 KHz the required match shall be ± 2.0 dB for gain and ± 10° for phase. The measured data, the computer model, and curves obtained from the computer model shall be submitted to the APSP-QAA with no restrictions on use by system manufacturers needing to insure system stability.



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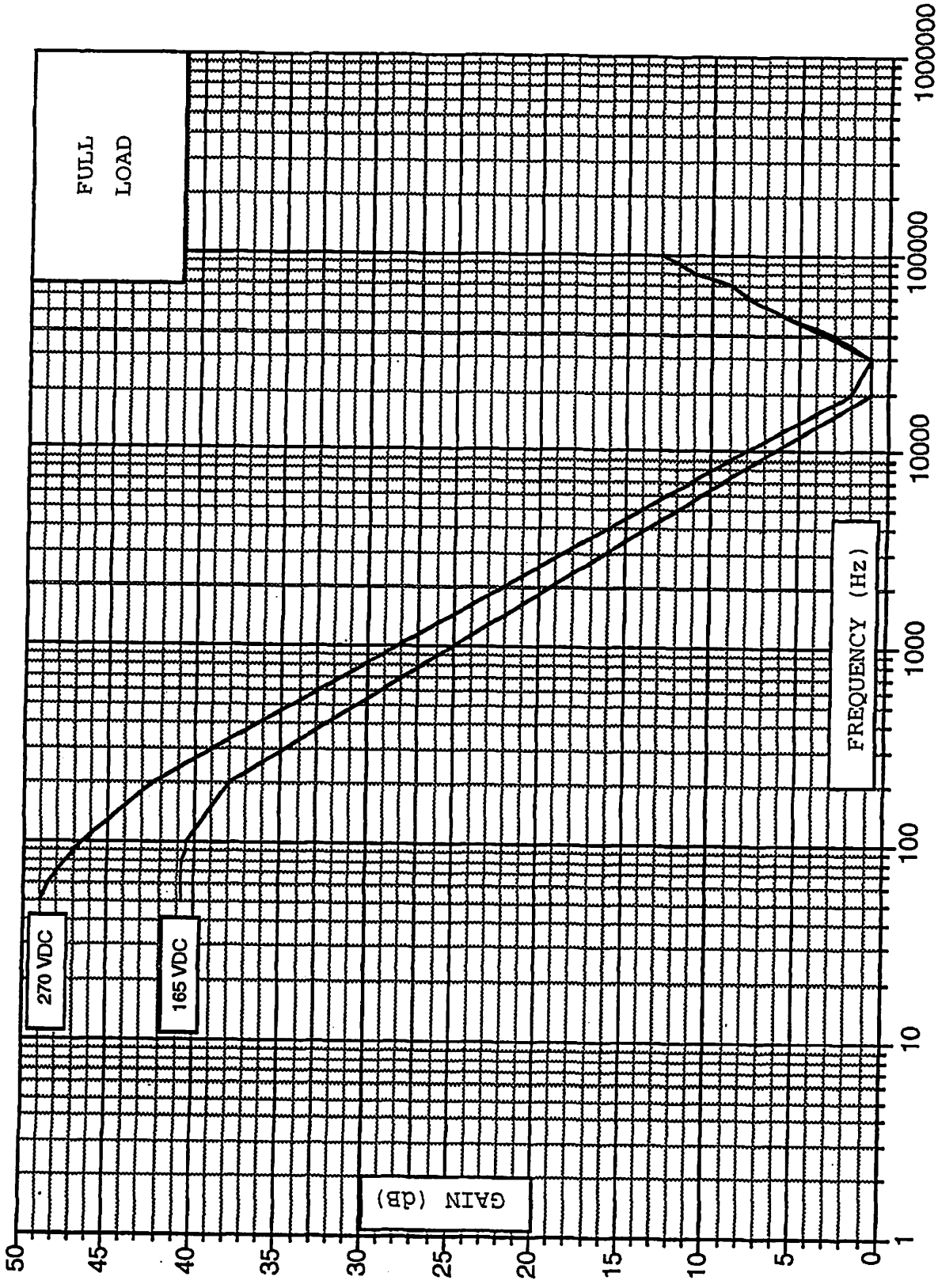



FIGURE 2. Minimum input impedance.



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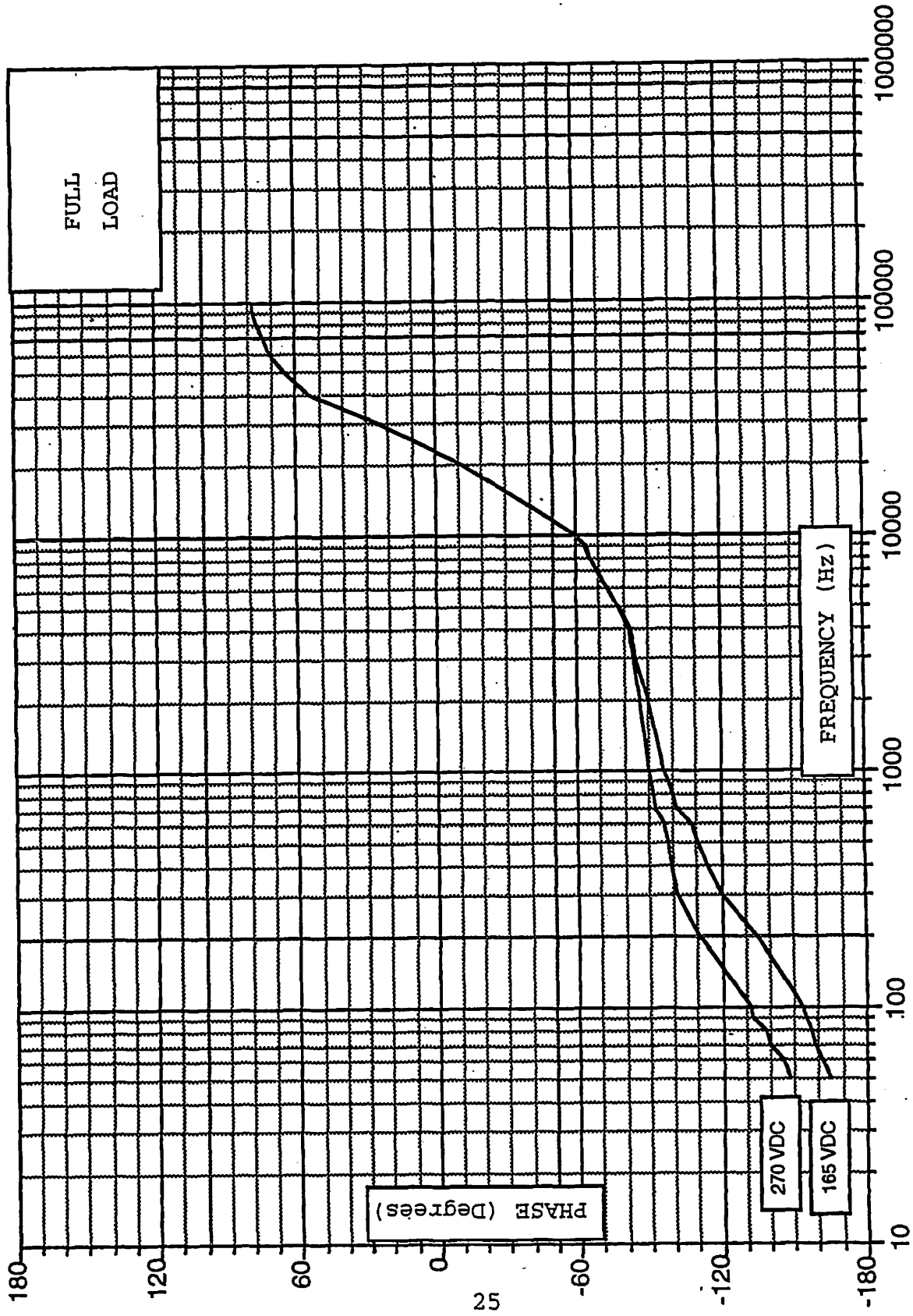



FIGURE 3. Minimum input impedance phase.



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3.8 Keying. The power supply key code shall be A88UG in accordance with MIL-P-29590.

3.9 Qualification. Initial qualification and periodic check shall be in accordance with MIL-P-29590 (see 4.5).

3.10 Quality conformance tests. All production power supplies shall be tested in accordance with 4.2 and footnotes 1 and 2 in table I at a rib temperature of 25 ± 5°C.

3.11 End-of-life limits. Power supplies which have been plugged into an operating system, or exceed 2 years from the date of manufacture and are to be tested, shall be capable of conforming to the 25°C end of life limits in table I (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-P-29590 shall apply.

4.2 Quality conformance tests. Quality conformance tests of all power supplies shall be in accordance with the procedures and equipment specified in MIL-P-29590.

4.3 Test equipment. Test equipment specified herein with a type designator shall be as specified in MIL-STD-1665. Other equipment or components shall be as specified on the figures or in the tables, and as specified below. Equivalent equipment may be used for quality conformance and qualification inspection testing provided the correlation requirements are met.

| ITEM | DESCRIPTION |
|---|---|
| 1 | DC power supply, 600 volts, 3 amperes, Sorenson DCR600-3B |
| 2 | DC power supply, 250 volts, 2.5 amperes, Kikusui PAD 250-2.5L |
| 3 | DC power supply, Harrison 6202B |
| 4 | DC power supply, Kikusui PAB 350-0.1A |
| 5 | Multimeters 1,2,3: Keithley 175 multimeter |
| 6 | Electronic load, 300 watt, Hewlett-Packard 6060A |
| 7 | Oscilloscope, storage, differential comparator amplifier, Tektronix 11402. |
| 8 | Current probe amplifier, Tektronix AM503 |
| 9 | Current probe, 20 ampere, Tektronix A6302 |
| 10 | Current probe, 100 ampere, Tektronix P6303 |
| 11 | Pulse generator, Wavetek 21 function generator |
| 12 | Pulse generator, Wavetek 187 function generator |
| 13 | Capacitor analyzer |



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4.4 Test procedures. The power supply shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the APSP-QAA and they conform to the requirements of MIL-P-29590.

4.4.2 Order of testing. Test conditions A through GG shall be performed in any order with the following exceptions. If test conditions A and BB are part of a sequence of tests, the order of testing shall be A then BB where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-P-29590 with the modifications or additions specified in 4.5.1 through 4.5.6.

4.5.1 Operating temperature and altitude tests. The operating temperature tests shall be conducted in accordance with MIL-P-29590. Before starting these tests, test conditions A through GG in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25 ± 5°C rib temperature. At turn on, following one hour stabilization at the low temperature, test conditions C1, C2, D2, D3, and P5 shall be performed. When electrical testing is called for at the end of each temperature (except 105°C) stabilization period, test conditions C, D, G1-G2, N1-N3, P1-P3, Q1-Q3, Q7, Q8, S1-S10, V1-V2, GG1, and GG2 shall be performed in the order specified in 4.4.2 using initial limits for the applicable operating temperature. When electrical testing is called for at 105°C, test conditions C3-C4, D1-D4, N1-N3, and Q1-Q3 shall be performed using the high temperature end-of-life limits. At all other times during the operating temperature tests, the power supply shall be operated in the test circuit shown on figure 4.

4.5.1.1 Barometric pressure. The barometric pressure tests shall be conducted in accordance with MIL-P-29590 except as modified herein. The power supply shall be tested in accordance with MIL-STD-202, method 105, test condition C.

4.5.2 Component temperature tests. The component temperatures shall be measured in accordance with MIL-P-29590. The power supply shall be operated as shown on figure 3. These measurements shall be performed on each initial qualification submission that is the result of a new vendor or of a major



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redesign of a previously qualified vendor. In addition to analyzing component temperatures, operation of the excessive temperature indicator and unsafe temperature indicator shall be verified to be within the temperature limits specified in table I.

4.5.3. Vibration tests. The vibration tests shall be conducted in accordance with MIL-P-29590. During the vibration tests, the power supply shall be operated in the circuit shown on figure 5. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25 ± 5°C rib temperature.

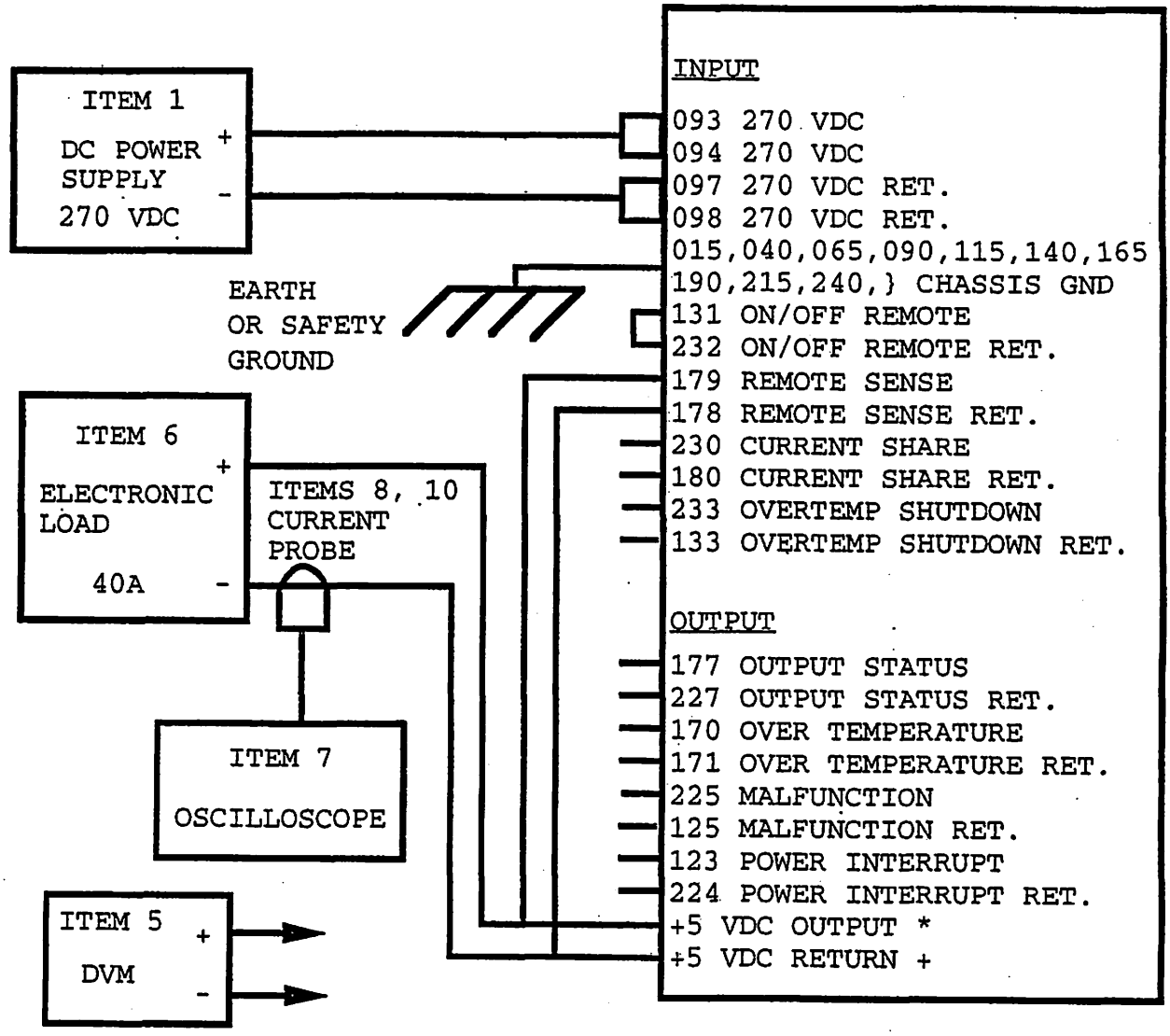
4.5.4 Shock tests. The mechanical shock tests shall be conducted in accordance with MIL-P-29590. During the shock tests, the power supply shall be operated in the circuit shown on figure 5. Following the shock tests, the electrical tests in table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25 ± 5°C rib temperature.

4.5.5. Long term stability test. The long term stability test shall be conducted in accordance with MIL-P-29590. The power supply shall be operated as shown on figure 4. After operating for 20 hours, the baseline output voltage readings shall be taken. At the conclusion of the test, prior to disconnecting the power supply or reducing the temperature, the output voltage readings shall again be recorded. The end-of-test voltages shall not change by more than 0.2 percent from the baseline voltage. Following the return of the power supply to 25°C, electrical tests A through V, X, Z, EE1, GG1, and GG2 shall be performed in the order specified in 4.4.2 using the 25 ± 5°C rib temperature end-of-life requirements in table I.

4.5.6 EMI test. The EMI tests shall be conducted in accordance with MIL-P-29590 and the procedures of MIL-STD-462. The loading shall be 90 to 100 percent of full load.



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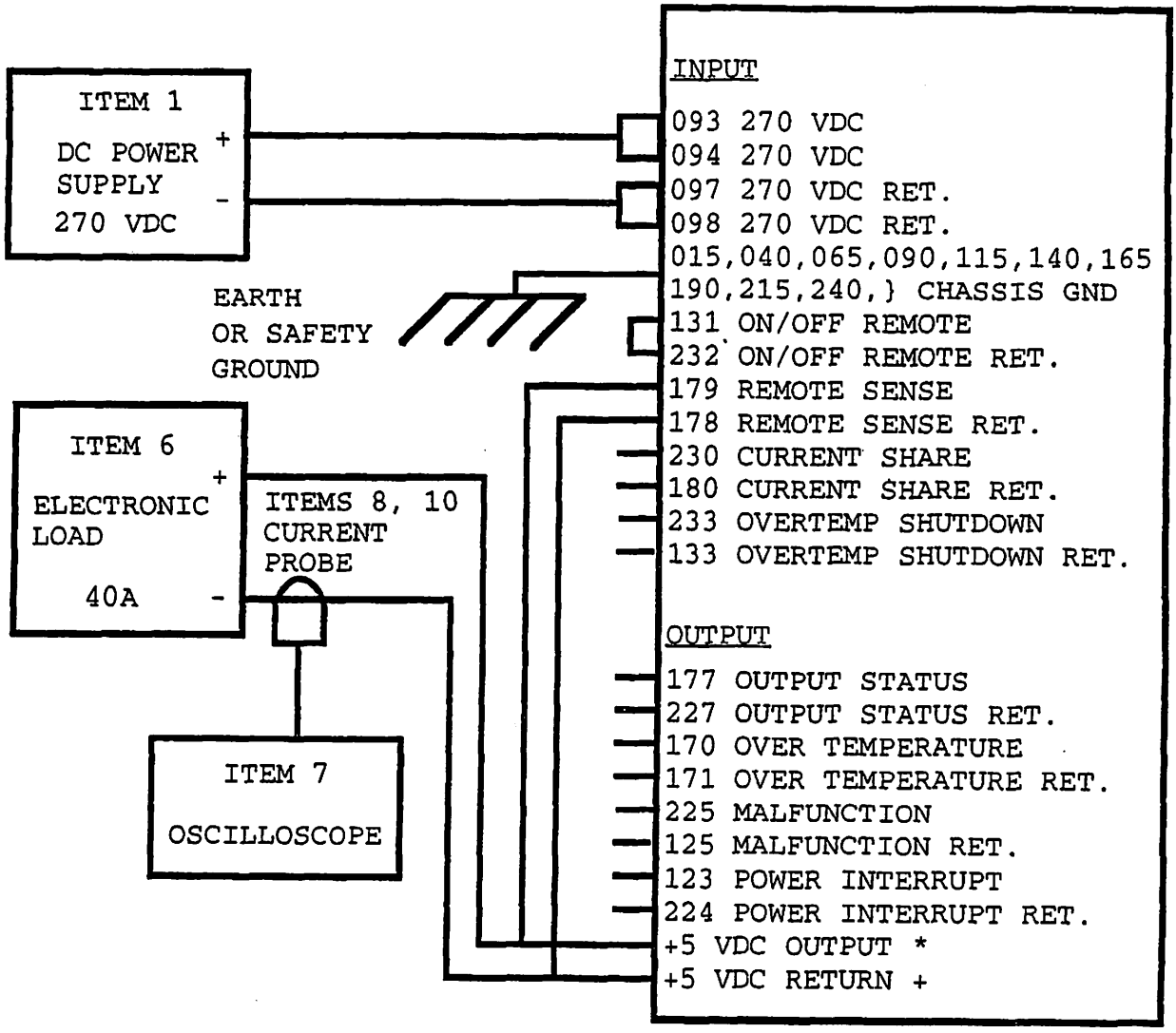

* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 4. Operating temperature, component temperature, and long term stability test circuit.



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* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 5. Vibration and shock test circuit.



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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-P-29590.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Power supplies specified herein are intended for use in military systems and subsystems.

6.2 Ordering data. Acquisition documents should specify the ordering data required in MIL-P-29590.

6.3 End-of-life. When any individual power supply parameter fails to conform to the end-of-life requirements in table I, the power supply is considered to have completed its useful life (see 3.11).

6.4 Power supply description.

Key code: A88UG

Power supply size: MIL-STD-1389, Format E

Mean time between failures (MTBF): 18,000 hours

Input voltage: 270 VDC derived from full wave rectified and filtered 3 phase, 400 Hz, 115 VAC power.

Output: Provides the following output voltage and current combination at any one time:

(a) Single output: 5.00 volts at 4 to 40 amperes.

6.5 Power supply function.

6.5.1 Stability with remote sense. Power supply stability may be adversely affected by certain load lead characteristics when the output voltage is remotely sensed. The inductance (L) of the load leads between the power supply output and the remote sense point together with the load capacitance (C) form an additional LC filter on the output of the power supply. When the corner frequency of this LC filter falls within the unity gain bandwidth of the power supply feedback loop, then the power supply phase margin is reduced. These conditions tend toward and can



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result in power supply instability. Caution should be exercised when utilizing remote sensing and having significant load lead inductance (L) and load capacitance (C). Power supply stability should be verified by analysis and also by dynamic load testing in the actual circuit configuration.

6.5.2 Output current rating. The output is rated at 4 to 40 amperes. For operation between 0 and 4 amperes, the load regulation and ripple limits are twice the limits for operation within the rated limits.

6.6 Recovery from overvoltage. The output voltage is not required to recover automatically from an output overvoltage fault and may require the input voltage to be taken to zero and then reapplied to reset the output. Cycling of the remote ON/OFF control will not accomplish this reset function.



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TABLE III. Isolation tests.

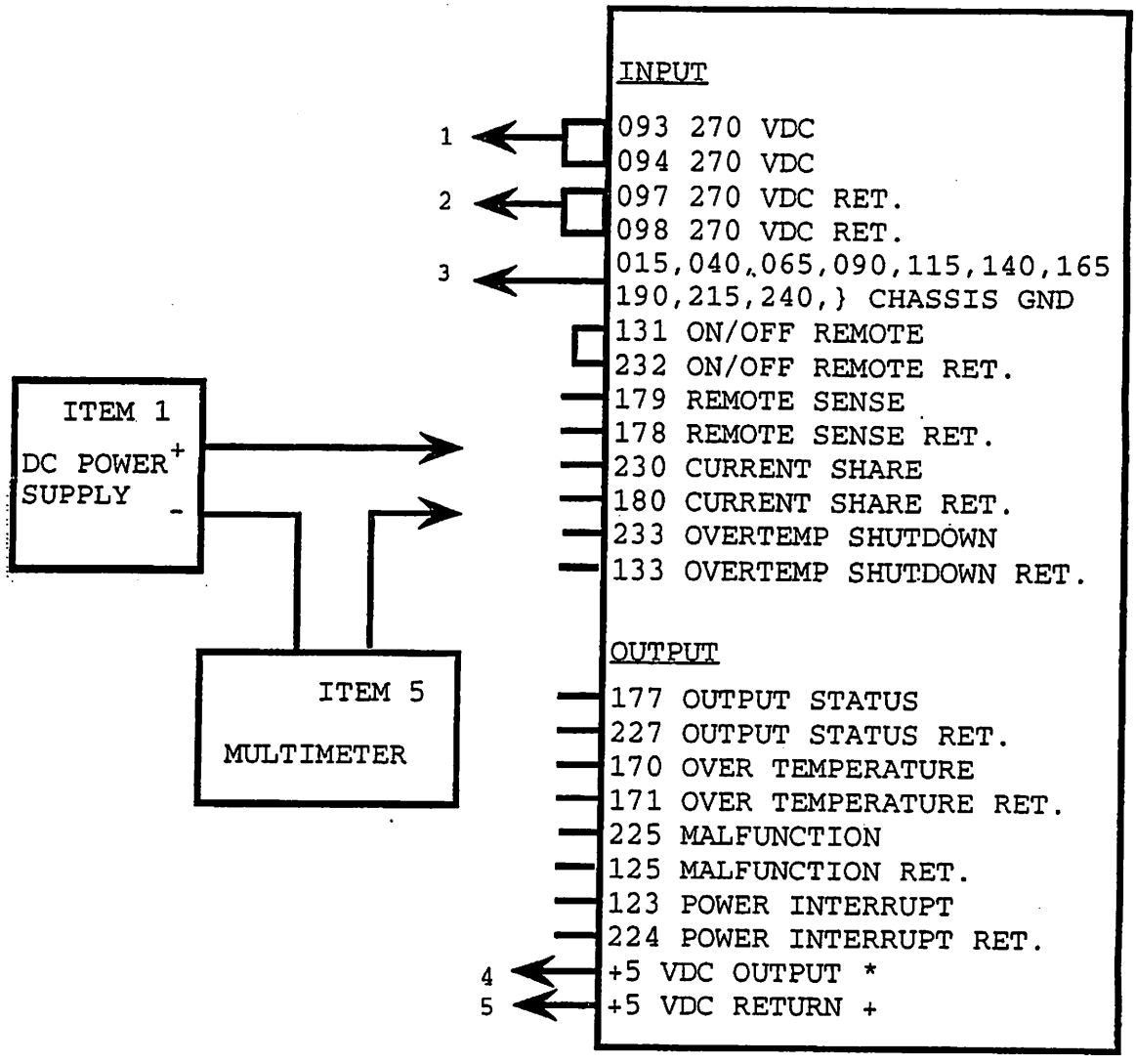
| Test cond | Parameter | Item 1 voltage ±10 percent | Connect item 5 | | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|
| | | | + Terminal | - Terminal | Min | Max | | | |
| A1 | Isolation -input to chassis | 600 VDC | 1 | 3 | 10 | INF | MOhm | 6 | X |
| A2 | Isolation -input to chassis | 600 VDC | 2 | 3 | 10 | INF | MOhm | 6 | X |
| A3 | Isolation -input to output | 600 VDC | 1 | 4 | 10 | INF | MOhm | 6 | X |
| A4 | Isolation -input to output | 600 VDC | 1 | 5 | 10 | INF | MOhm | 6 | X |
| A5 | Isolation -input to output | 600 VDC | 2 | 4 | 10 | INF | MOhm | 6 | X |
| A6 | Isolation -input to output | 600 VDC | 2 | 5 | 10 | INF | MOhm | 6 | X |

NOTE:

1. Isolation (MOhms) = $\frac{\text{Item 1 (volts)}}{\text{Item 5 (}\mu\text{ADC)}}$



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* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 6. Isolation test circuit for table III.



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TABLE IV. Continuity tests.

| Test cond | Parameter | Connect item 5 | | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|
| | | + Terminal | - Terminal | Min | Max | | | |
| B1 | Chassis pin to chassis continuity | 040 | Frame | 0.0 | 1.0 | Ohm | - | X |
| B2 | 270V pin continuity | 093 | 094 | 0.0 | 1.0 | Ohm | - | X |
| B3 | Chassis GND continuity | 040 | 090 | 0.0 | 1.0 | Ohm | - | X |
| B4 | Chassis GND continuity | 040 | 115 | 0.0 | 1.0 | Ohm | - | X |
| B5 | Chassis GND continuity | 040 | 140 | 0.0 | 1.0 | Ohm | - | X |
| B6 | Chassis GND continuity | 040 | 190 | 0.0 | 1.0 | Ohm | - | X |
| B7 | Chassis GND continuity | 040 | 165 | 0.0 | 1.0 | Ohm | - | X |
| B8 | Chassis GND continuity | 040 | 215 | 0.0 | 1.0 | Ohm | - | X |
| B9 | Chassis GND continuity | 040 | 240 | 0.0 | 1.0 | Ohm | - | X |
| B10 | Chassis GND continuity | 040 | 015 | 0.0 | 1.0 | Ohm | - | X |
| B11 | Chassis GND continuity | 040 | 065 | 0.0 | 1.0 | Ohm | - | X |
| B12 | 270V RET pin continuity | 097 | 098 | 0.0 | 1.0 | Ohm | - | X |
| B13 | +5V continuity | 001 | 002 | 0.0 | 1.0 | Ohm | - | X |
| B14 | +5V continuity | 001 | 003 | 0.0 | 1.0 | Ohm | - | X |
| B15 | +5V continuity | 001 | 004 | 0.0 | 1.0 | Ohm | - | X |
| B16 | +5V continuity | 001 | 005 | 0.0 | 1.0 | Ohm | - | X |
| B17 | +5V continuity | 001 | 006 | 0.0 | 1.0 | Ohm | - | X |
| B18 | +5V continuity | 001 | 007 | 0.0 | 1.0 | Ohm | - | X |



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TABLE IV. Continuity tests - Continued.

| Test cond | Parameter | Connect item 5 | | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|
| | | + Terminal | - Terminal | Min | Max | | | |
| B19 | +5V continuity | 001 | 008 | 0.0 | 1.0 | Ohm | - | X |
| B20 | +5V continuity | 001 | 009 | 0.0 | 1.0 | Ohm | - | X |
| B21 | +5V continuity | 001 | 051 | 0.0 | 1.0 | Ohm | - | X |
| B22 | +5V continuity | 001 | 052 | 0.0 | 1.0 | Ohm | - | X |
| B23 | +5V continuity | 001 | 053 | 0.0 | 1.0 | Ohm | - | X |
| B24 | +5V continuity | 001 | 054 | 0.0 | 1.0 | Ohm | - | X |
| B25 | +5V continuity | 001 | 055 | 0.0 | 1.0 | Ohm | - | X |
| B26 | +5V continuity | 001 | 056 | 0.0 | 1.0 | Ohm | - | X |
| B27 | +5V continuity | 001 | 057 | 0.0 | 1.0 | Ohm | - | X |
| B28 | +5V continuity | 001 | 058 | 0.0 | 1.0 | Ohm | - | X |
| B29 | +5V continuity | 001 | 059 | 0.0 | 1.0 | Ohm | - | X |
| B30 | +5V continuity | 001 | 102 | 0.0 | 1.0 | Ohm | - | X |
| B31 | +5V continuity | 001 | 104 | 0.0 | 1.0 | Ohm | - | X |
| B32 | +5V continuity | 001 | 106 | 0.0 | 1.0 | Ohm | - | X |
| B33 | +5V continuity | 001 | 108 | 0.0 | 1.0 | Ohm | - | X |
| B34 | +5V continuity | 001 | 110 | 0.0 | 1.0 | Ohm | - | X |
| B35 | +5V RET continuity | 016 | 017 thru 028 | 0.0 | 1.0 | Ohm | - | X |
| B36 | +5V RET continuity | 016 | 066 thru 0078 | 0.0 | 1.0 | Ohm | - | X |



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TABLE IV. Continuity tests - Continued.

| Test cond | Parameter | Connect item 5 | | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|
| | | + Terminal | - Terminal | Min | Max | | | |
| B37 | +5V RET continuity | 016 | 116 | 0.0 | 1.0 | Ohm | - | X |
| B38 | +5V RET continuity | 016 | 118 | 0.0 | 1.0 | Ohm | - | X |
| B39 | +5V RET continuity | 016 | 120 | 0.0 | 1.0 | Ohm | - | X |
| B40 | +5V RET continuity | 016 | 122 | 0.0 | 1.0 | Ohm | - | X |
| B41 | +5V RET continuity | 016 | 124 | 0.0 | 1.0 | Ohm | - | X |
| B42 | +5V RET continuity | 016 | 126 | 0.0 | 1.0 | Ohm | - | X |
| B43 | +5V RET continuity | 016 | 128 | 0.0 | 1.0 | Ohm | - | X |
| B44 | +5V RET continuity | 010 | 011 thru 014 | 0.0 | 1.0 | Ohm | - | X |
| B45 | +5V RET continuity | 010 | 060 thru 064 | 0.0 | 1.0 | Ohm | - | X |
| B46 | +5V RET continuity | 010 | 110 | 0.0 | 1.0 | Ohm | - | X |
| B47 | +5V RET continuity | 010 | 112 | 0.0 | 1.0 | Ohm | - | X |
| B48 | +5V RET continuity | 010 | 114 | 0.0 | 1.0 | Ohm | - | X |
| B49 | +5V continuity | 029 | 030 thru 037 | 0.0 | 1.0 | Ohm | - | X |
| B50 | +5V continuity | 029 | 079 thru 087 | 0.0 | 1.0 | Ohm | - | X |
| B51 | +5V continuity | 029 | 130 | 0.0 | 1.0 | Ohm | - | X |
| B52 | +5V continuity | 029 | 132 | 0.0 | 1.0 | Ohm | - | X |
| B53 | +5V continuity | 029 | 134 | 0.0 | 1.0 | Ohm | - | X |
| B54 | +5V continuity | 029 | 136 | 0.0 | 1.0 | Ohm | - | X |



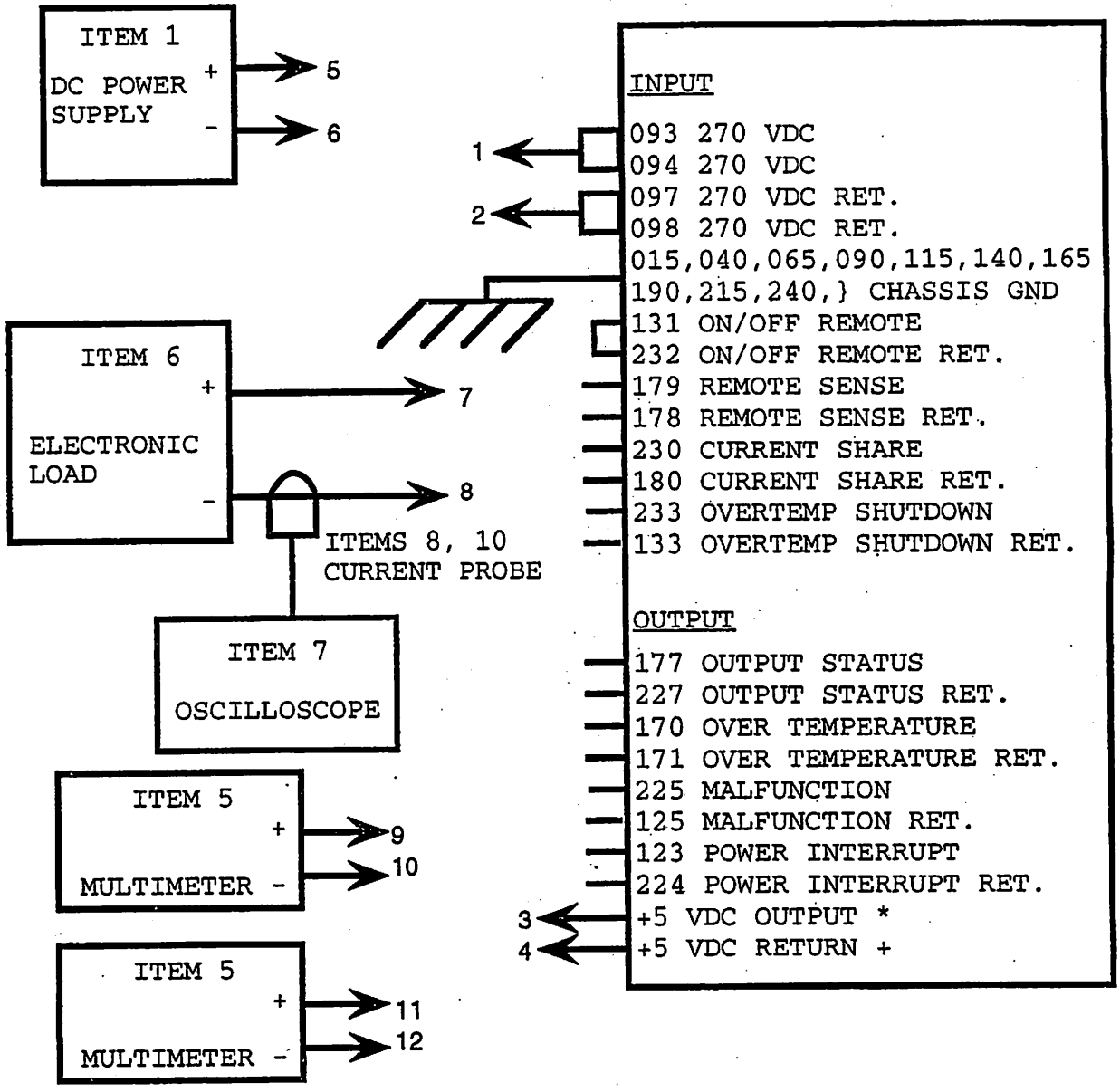
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE V. Load regulation tests.

| Test cond | Parameter | Item 1 +0/ -2 per-cent | Item 6 +0/ -2 per-cent | Connect | | Connect | | Initial limits | | Unit | Test fig | test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 1 to | 2 to | 3 to | 4 to | Min | Max | | | |
| C1 | Load regulation | 270V | 40A | 5, 11 | 6, 12 | 7, 9 | 8, 10 | 4.950 | 5.050 | VDC | 7 | X |
| C2 | Load regulation | 270V | 4A | 5, 11 | 6, 12 | 7, 9 | 8, 10 | 4.950 | 5.050 | VDC | 7 | X |
| C3 | Load regulation | 270V | <4A | 5, 11 | 6, 12 | 7, 9 | 8, 10 | 4.900 | 5.100 | VDC | 7 | X |
| C4 | Load regulation | 270V | 0A | 5, 11 | 6, 12 | 7, 9 | 8, 10 | 4.900 | 5.100 | VDC | 7 | X |



Source: https://assist.dla.mil -- Downloaded: 202640-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 7. Load regulation test circuit for table V.



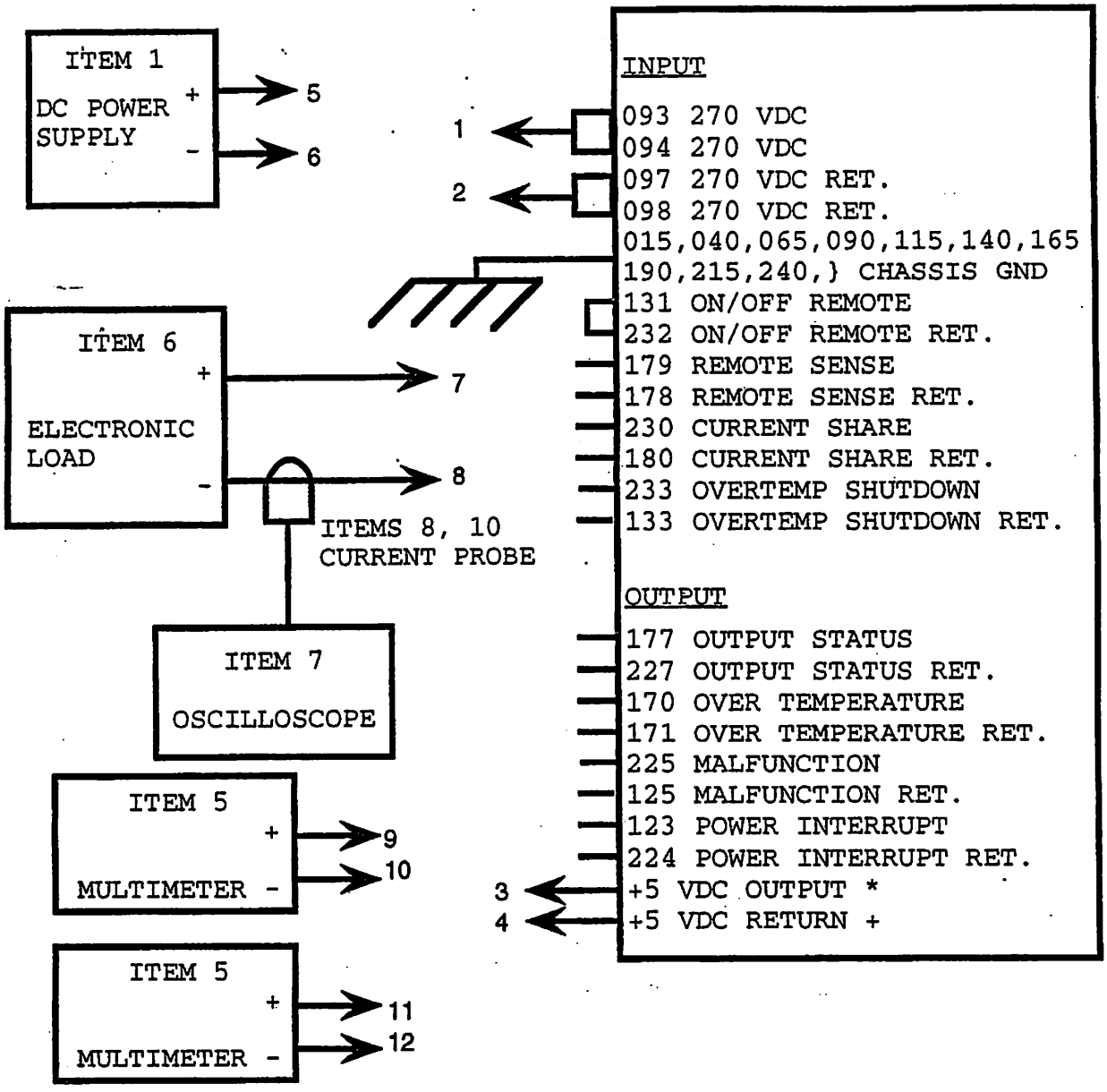
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE VI. Line regulation.

| Test cond | Parameter | Item 6 current +0/-2 percent | Connect | | Connect | | Input voltage item 1 | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 1 to | 2 to | 3 to | 4 to | | Min | Max | | | |
| D1 | Line regulation | 4A | 5, 11 | 6, 12 | 7, 9 | 8, 10 | 240V | 4.950 | 5.050 | VDC | 8 | X |
| D2 | Line regulation | 40A | 5, 11 | 6, 12 | 7, 9 | 8, 10 | 240V | 4.950 | 5.050 | VDC | 8 | X |
| D3 | Line regulation | 4A | 5, 11 | 6, 12 | 7, 9 | 8, 10 | 280V | 4.950 | 5.050 | VDC | 8 | X |
| D4 | Line regulation | 40A | 5, 11 | 6, 12 | 7, 9 | 8, 10 | 280V | 4.950 | 5.050 | VDC | 8 | X |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 8. Line regulation test circuit for table VI.



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TABLE VII. Low input voltage protection.

| Test cond | Parameter | Item 6 +0, -2 percent | Connect | | Item 1 +0/ -2 percent | Item 7 +0/ -2 percent | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 1 to | 2 to | | | | Min | Max | | | |
| E1 | Low input voltage protection | 40A | 3,5 | 4,6 | 270V to 240V | Record IIN1, IIN2 | IIN ratio | -- | 1.8 | - | 9 | Q |

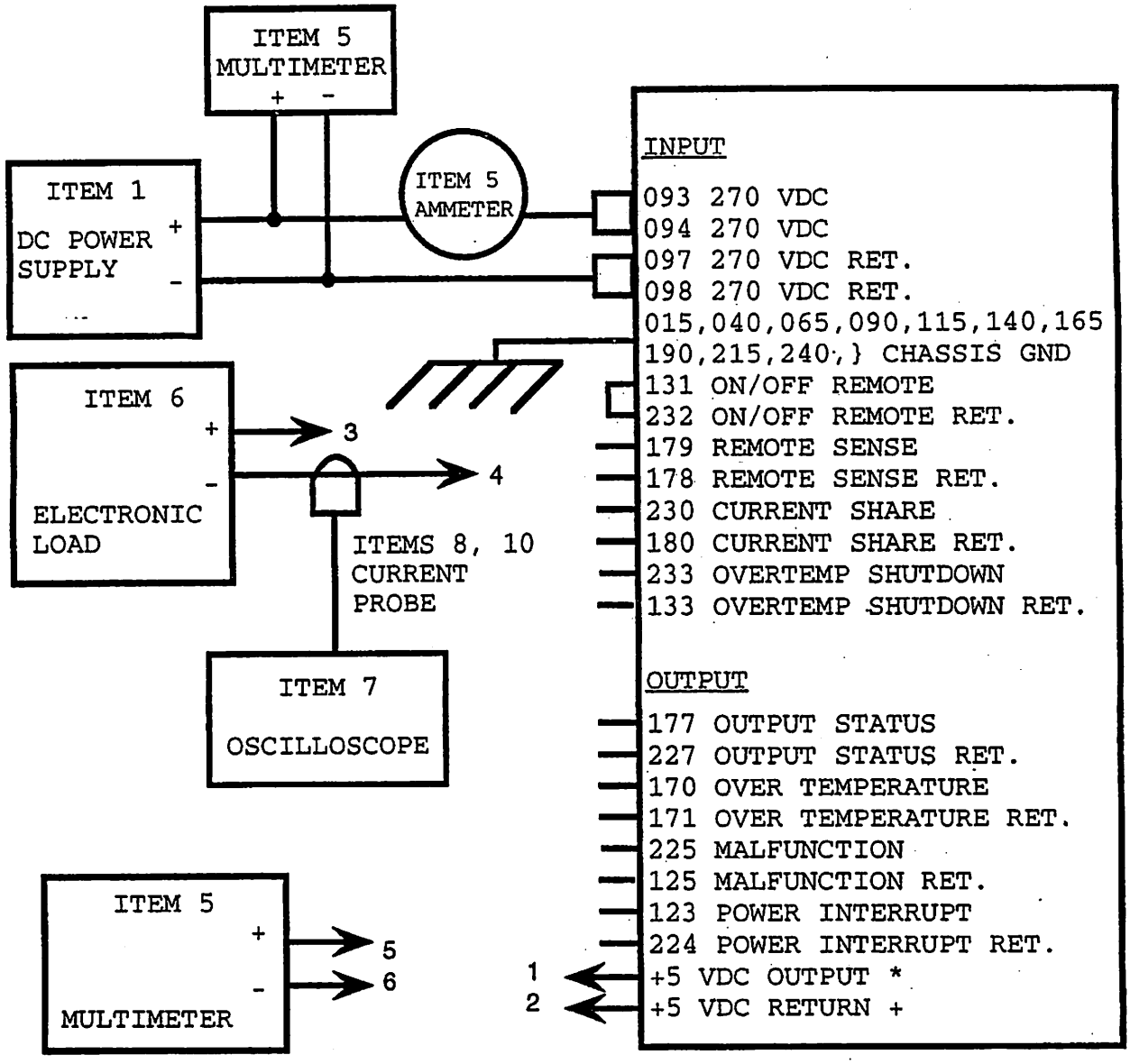
NOTE:

1. IIN1 shall be measured with Item 1 set to 270 VDC. IIN2 shall be measured with Item 1 set to 240 VDC. IIN ratio is calculated as follows:

IIN ratio = IIN2/IIN1.



Source: https://assist.dla.mil -- Downloaded: 202610-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 9. Low input voltage protection for table VII.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

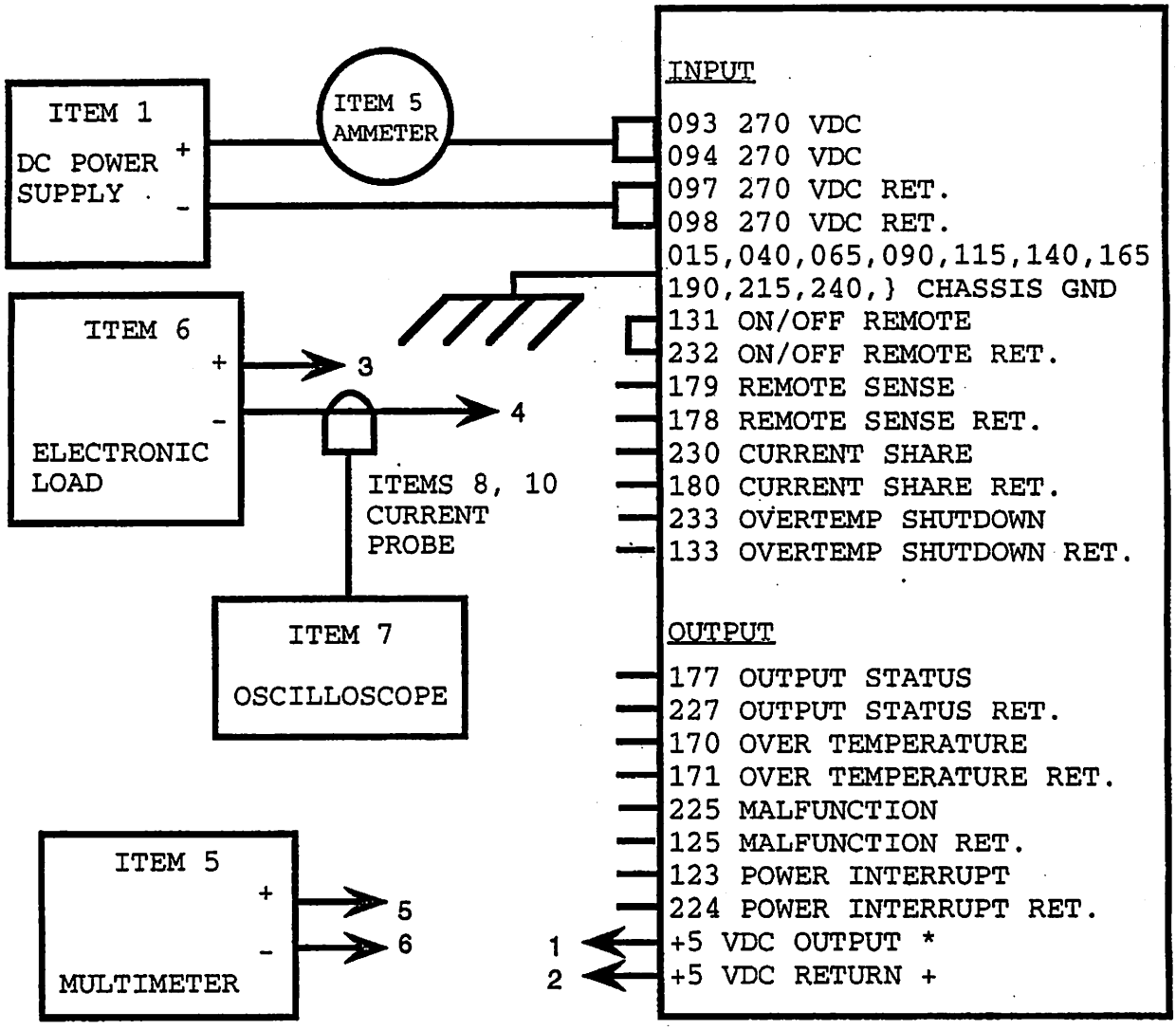
TABLE VIII. Efficiency.

| Test cond | Parameter | Connect | | Item 1 +0/ -2 percent | Item 6 +0/ -2 percent | Meas % E | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | Min | Max | | | |
| G1 | % Efficiency | 3,5 | 4,6 | 270V | 40A | 1/ | 80 | -- | % | 10 | X |
| G2 | % Efficiency | 3,5 | 4,6 | 270V | 20A | 1/ | 80 | -- | % | 10 | Q |

1/ Percent efficiency = $\frac{(VOUT \times IOUT \times 100)}{(VIN \times IIN)}$



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 10. Efficiency test circuit for table VIII.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE IX. Inrush current.

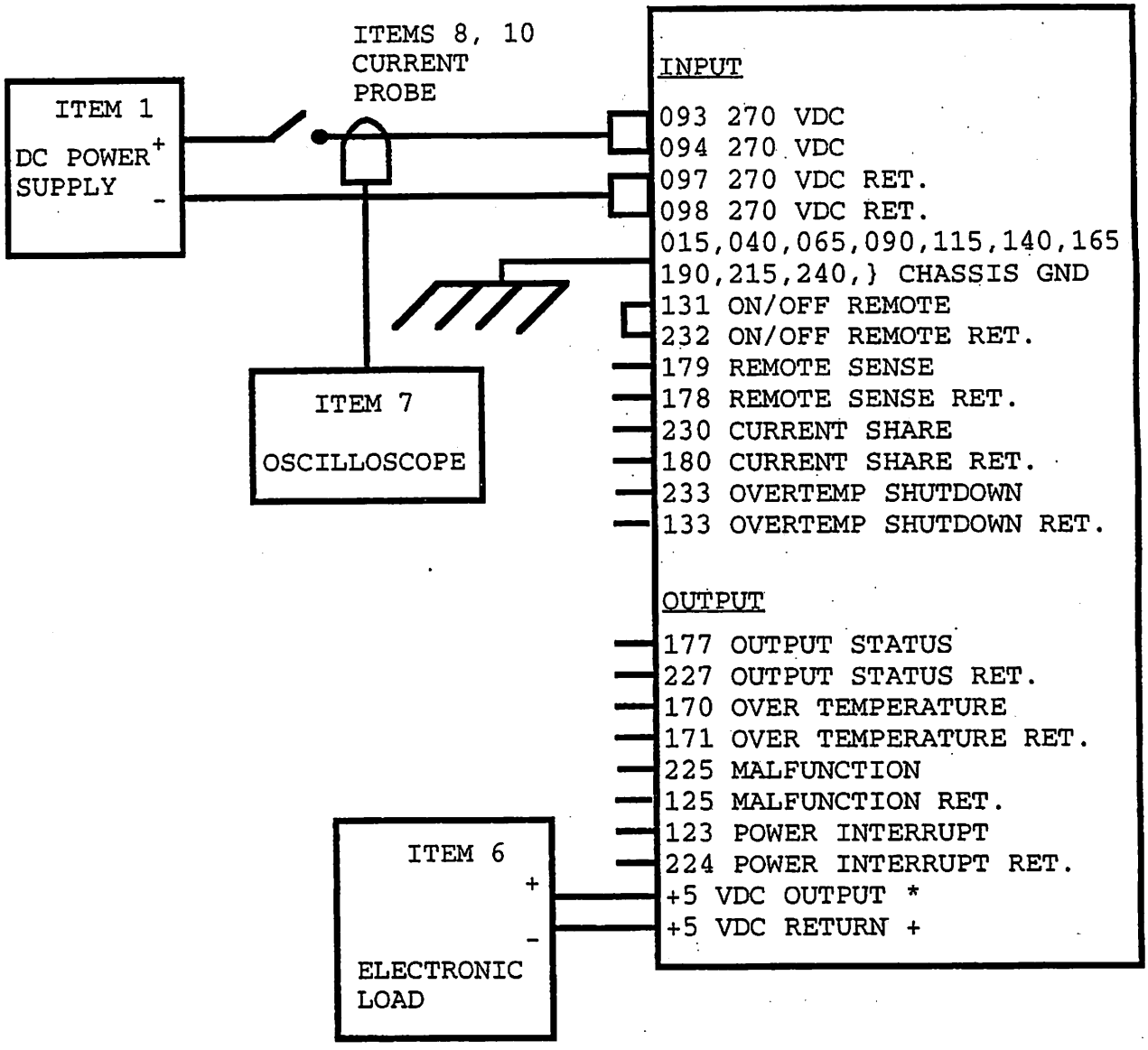
| Test cond | Parameter | Item 1 +0/-2 percent | Item 6 +0/-2 percent | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Min | Max | | | |
| K1 | Inrush current | 270V | 40A | IIN | - | 100 | Apk | 11 | Q |

NOTE:

1. Insure that the prime power source does not limit the inrush current to the power supply under test.



Source: https://assist.dla.mil -- Downloaded: 202610-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 11. Inrush current test circuit for table IX.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
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TABLE X. Remote sense compensation test.

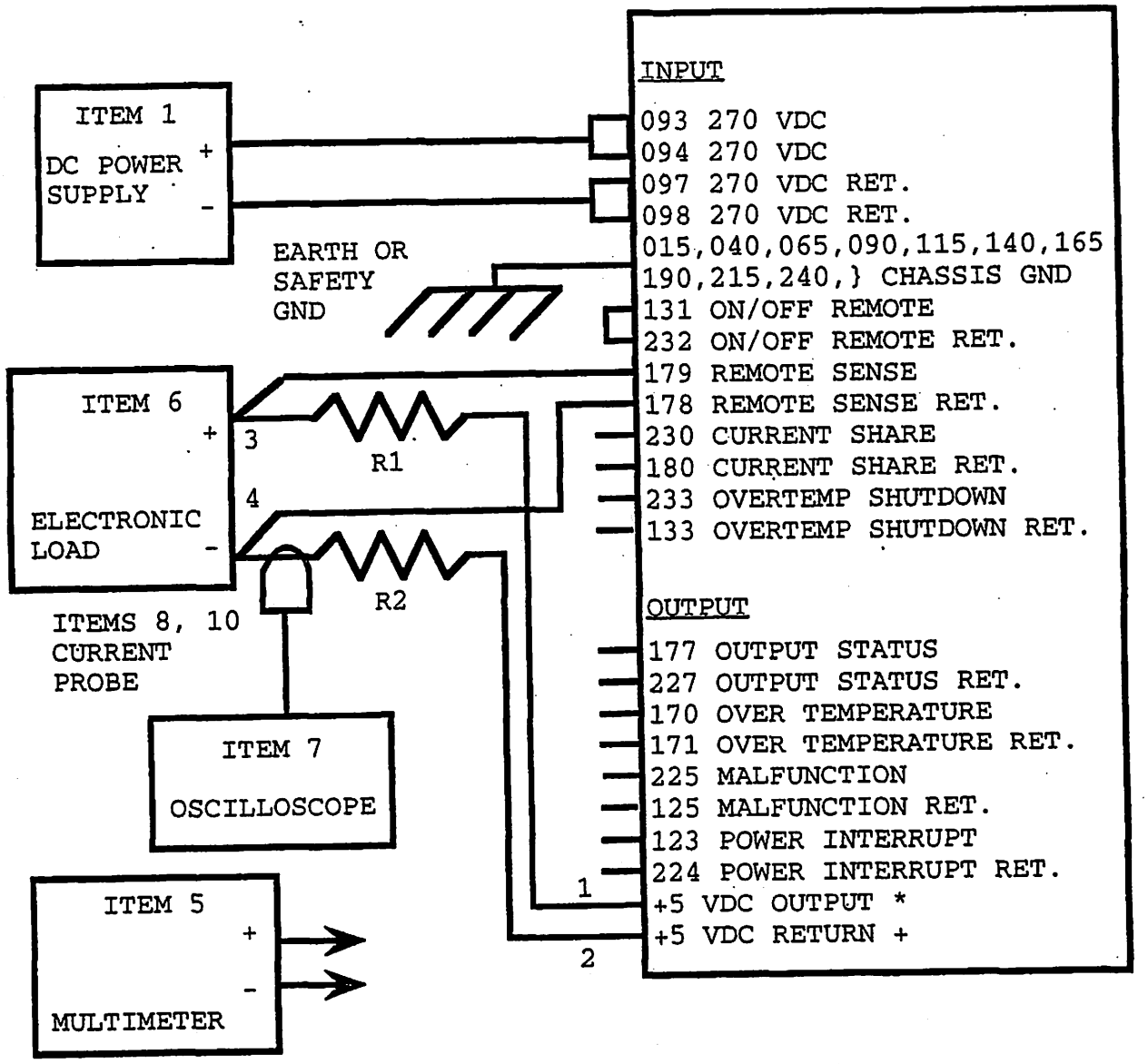
| Test cond | Parameter | Item 1 +0/-2 per-cent | Item 6 +0/-2 per-cent | Initial limits 25 ± 5°C mounting surface temperature | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | | | |
| L1 | Remote sense compensa-tion, 5.0V | 270V | 40A | 4.95 | 5.05 | VDC | 12 | Q |

NOTE:

1. R1 and R2 shall be the total resistance (including lead and contact resistance) to cause a 0.20V, +0.02, -0.00 volt drop when the voltage at 1 and 2 is compared to the voltage at 3 and 4.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 12. Remote sense compensation test circuit for table X.



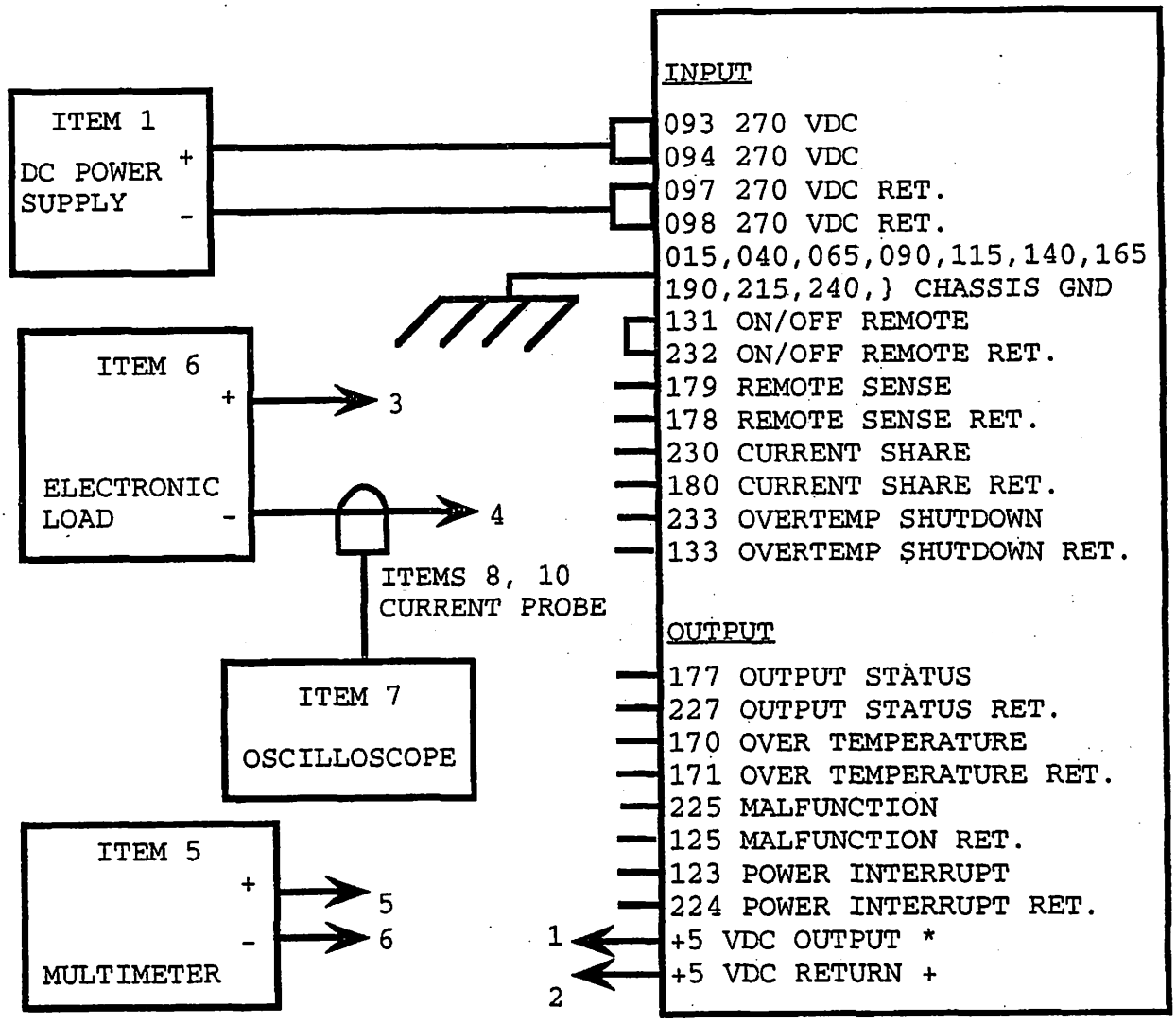
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XI. Overload protection.

| Test cond | Parameter | Connect | | Item 1 +0/ -2 percent | Item 6 +0/ -2 percent | Item 5 | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | | Min | Max | | | |
| N1 | Overload protection | 3,5 | 4,6 | 270V | 40A | | VOUT | 4.950 | 5.050 | VDC | 13 | X |
| N2 | Overload protection | 3,5 | 4,6 | 270V | | <4.895V | IOUT | 44 | 50 | ADC | 13 | X |



Source: https://assist.dla.mil -- Downloaded: 202610-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 13. Overload protection test circuit for table XI.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XII. Short circuit recovery.

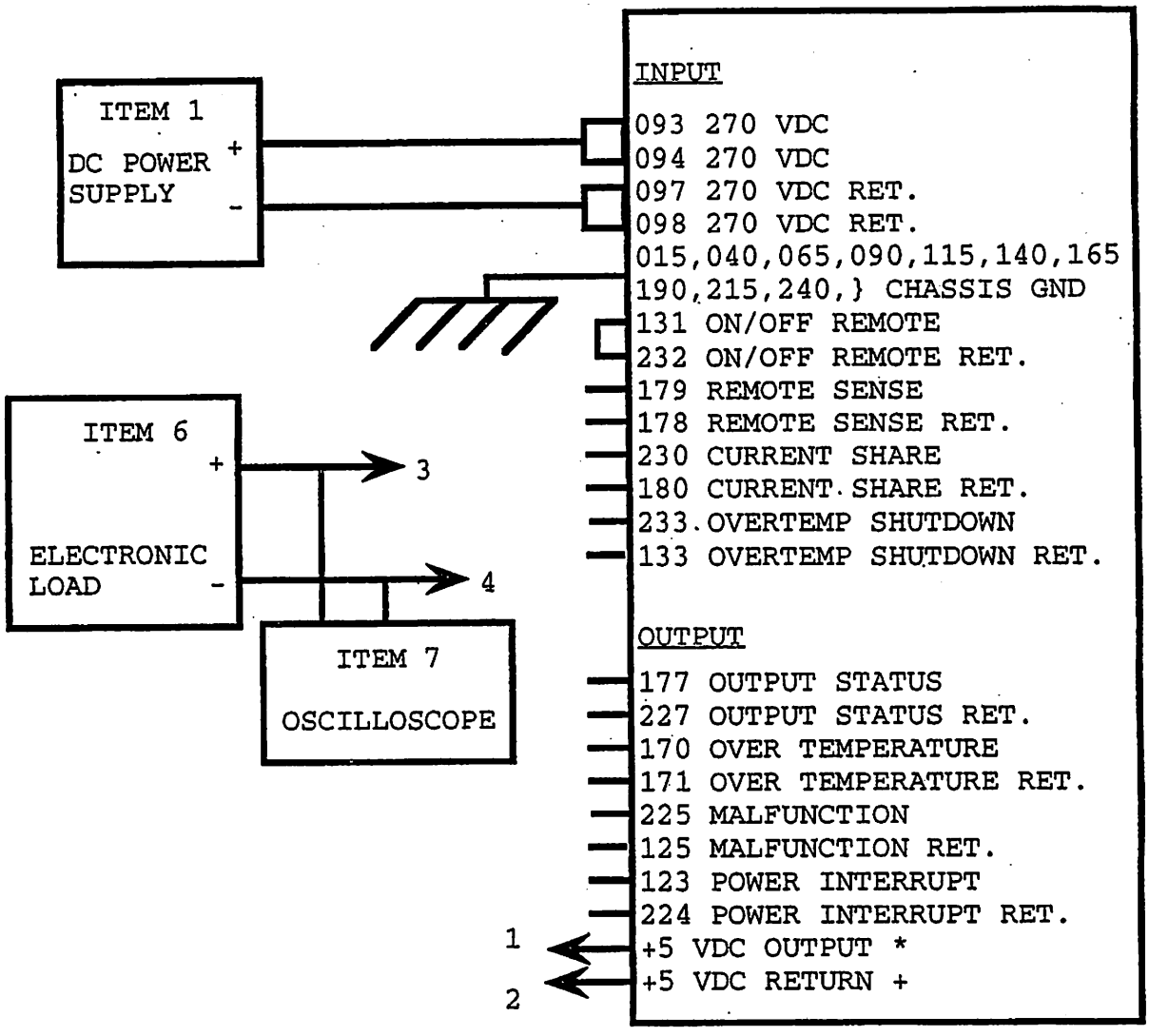
| Test cond | Parameter | Connect | | Item 1 +0/ -2 percent | Item 6 +0/ -2 percent | Item 7 | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | | Min | Max | | | |
| N3 | Short circuit recovery | 3 | 4 | 270V | 40A | 4.950V to 5.050V | tR1 | -- | 100 | msec | 14 | Q |

NOTE:

1. Depress the short circuit mode switch on the E load. Energize the power supply (item 1). Release the short circuit switch. Power supply shall recover within the specified time.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 14. Short circuit recovery test circuit for table XII.



Source: https://assist.dla.mil -- Downloaded: 2026-02-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XIII. Short circuit overshoot.

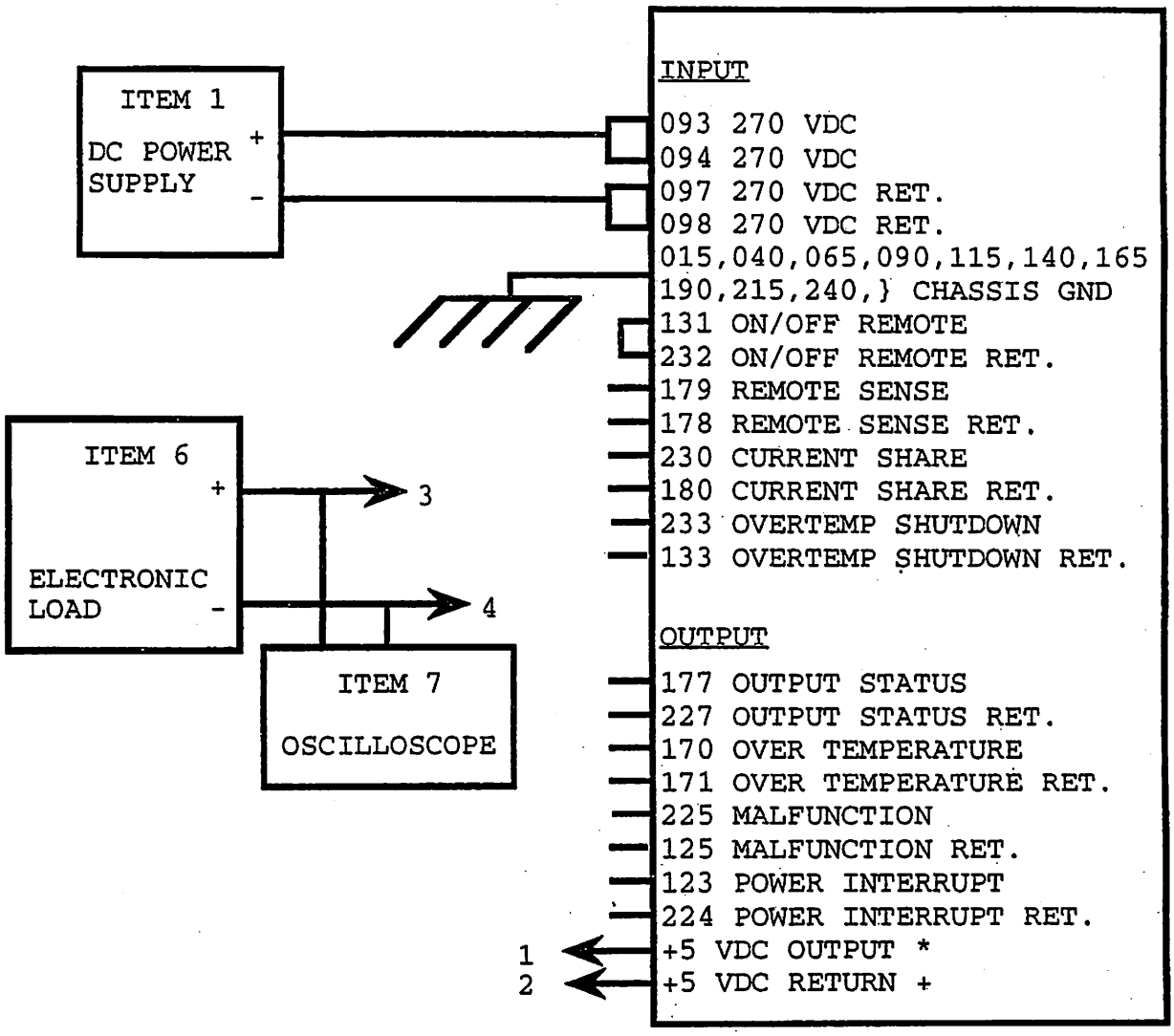
| Test cond | Parameter | Connect | | Item 1 +0/ -2 percent | Item 6 +0/ -2 percent | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | Min | Max | | | |
| N4 | Short circuit overshoot | 3 | 4 | 270V | 40A | Under/ Over Shoot | -0.25 | 0.25 | V | 15 | Q |

NOTE:

1. Output shall be shorted by depressing short circuit switch on the E load (item 6). When short circuit switch is released, the VOUT including overshoot/undershoot shall fall between specified limits.



Source: https://assist.dla.mil -- Downloaded: 202640-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 15. Short circuit overshoot test circuit for table XIII.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XIV. Short circuit current.

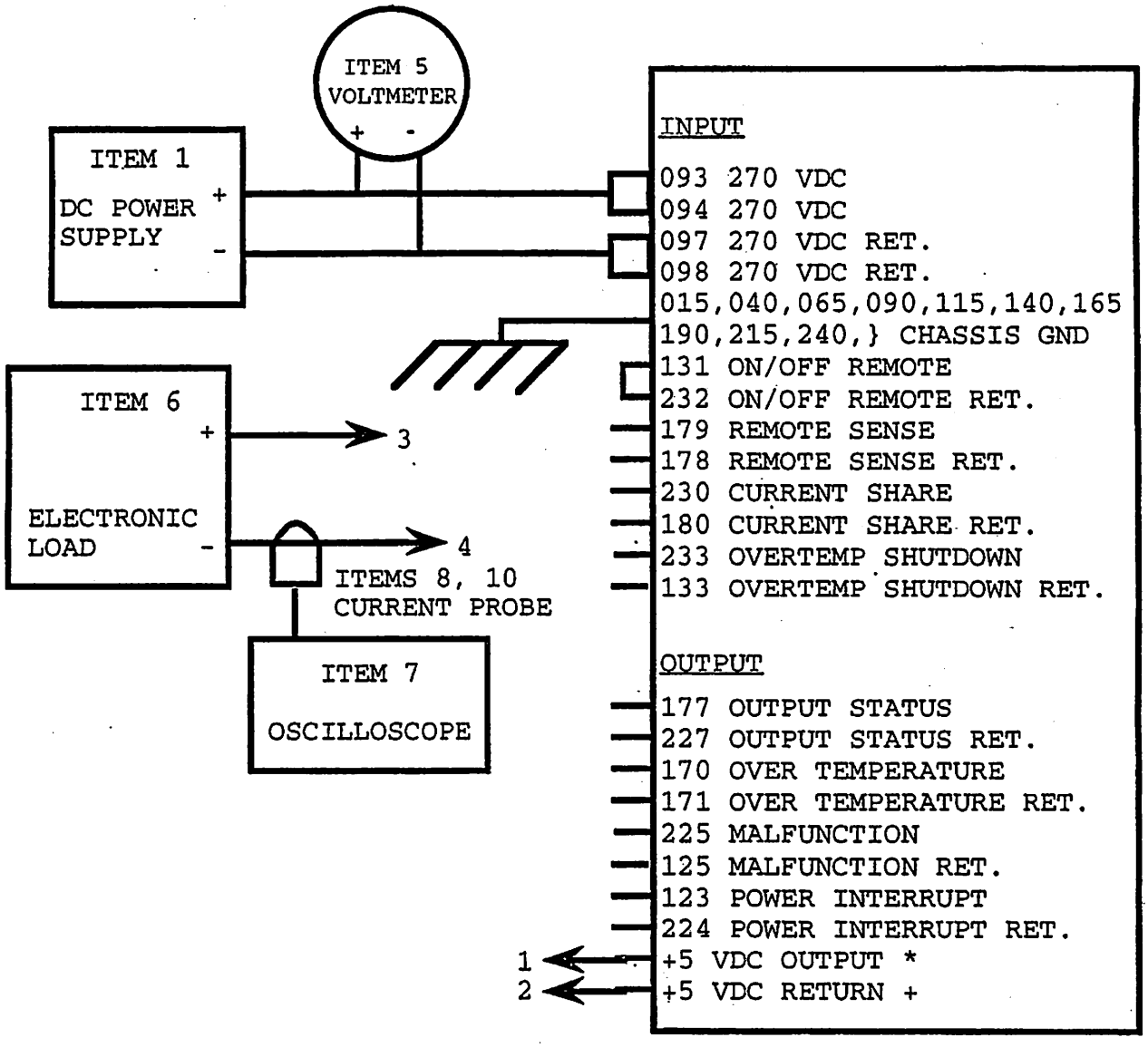
| Test cond | Parameter | Connect | | Item 1 +0/-2 percent | Item 6 1/ +0/-2 percent | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | Min | Max | | | |
| N5 | Short circuit current | 3 | 4 | 270V | 40A | IOUT | -- | 50 | ADC | 16 | S |

NOTE:

1. Electronic load shall be set to draw 40 amps. Short circuit mode switch shall then be depressed. Read short circuit current and then release short circuit switch.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 16. Short circuit current test circuit for table XIV.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XV. Turn-on transient time.

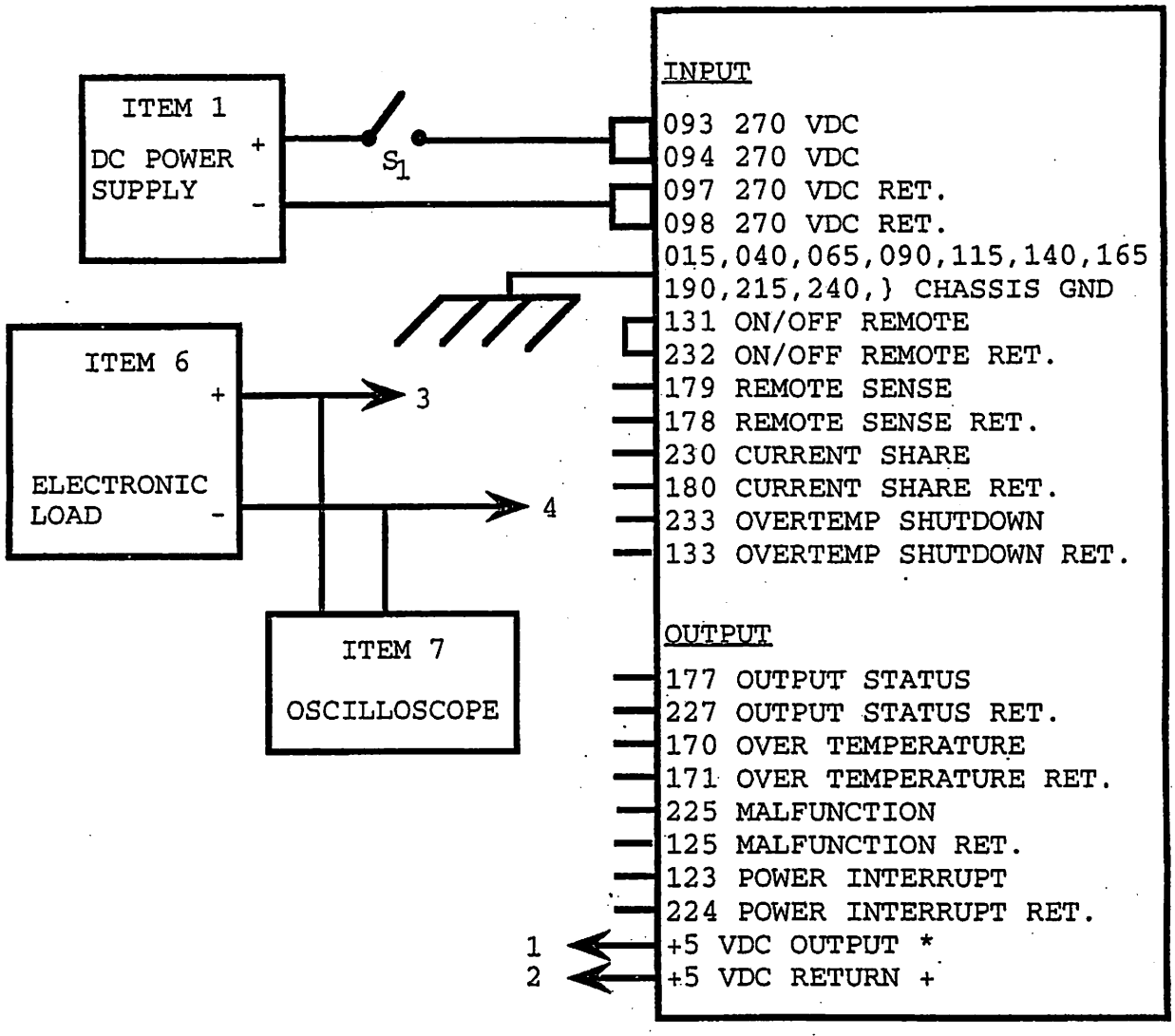
| Test cond | Parameter | Connect | | Item 1 1/ +0/ -2 percent | Item 6 +0/ -2 percent | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | Min | Max | | | |
| P1 | Turn-on transient | 3 | 4 | 270V | 40A | VOUT | -0.25 | 0.25 | V | 17 | Q |
| P2 | Turn-on time | 3 | 4 | 270V | 40A | t | -- | 250 | msec | 17 | S |

NOTE:

1. Switch $S_1$ shall be shut to supply power to the UUT.



Source: https://assist.dla.mil -- Downloaded: 202610-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 17. Turn-on time, transient test circuit for table XV.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XVI. Turn-off transient.

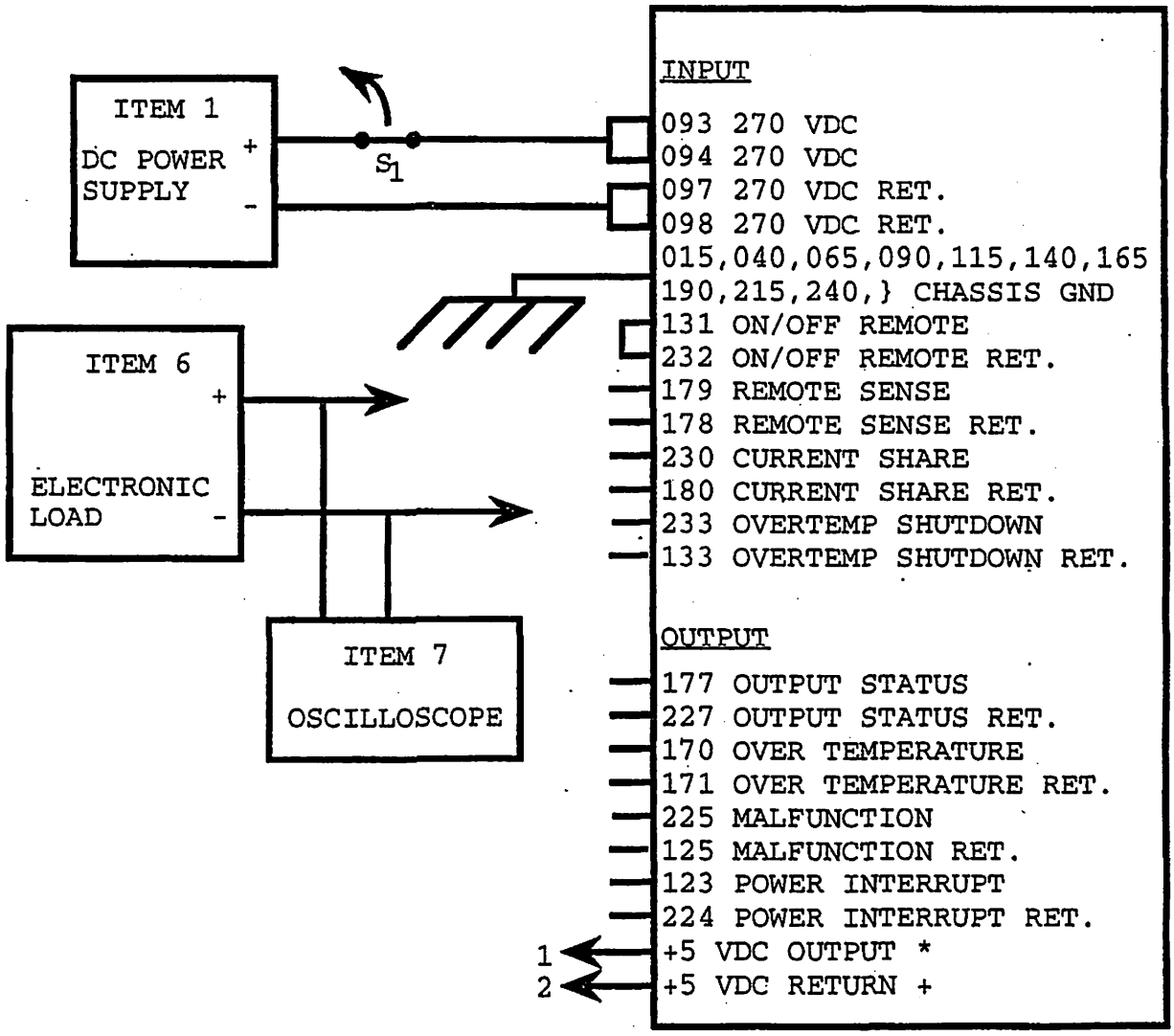
| Test cond | Parameter | Connect | | Item 1 +0/-2 percent | Item 6 +0/-2 percent | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | Min | Max | | | |
| P3 | Turn-off transient | 3 | 4 | 270V | 40A | VOUT | -0.25 | 0.25 | V | 18 | Q |

NOTE:

1. Switch S1 shall be opened to cause transient to occur on output.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 18. Turn-off transient test circuit for table XVI.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XVII. Control turn-on/off transient.

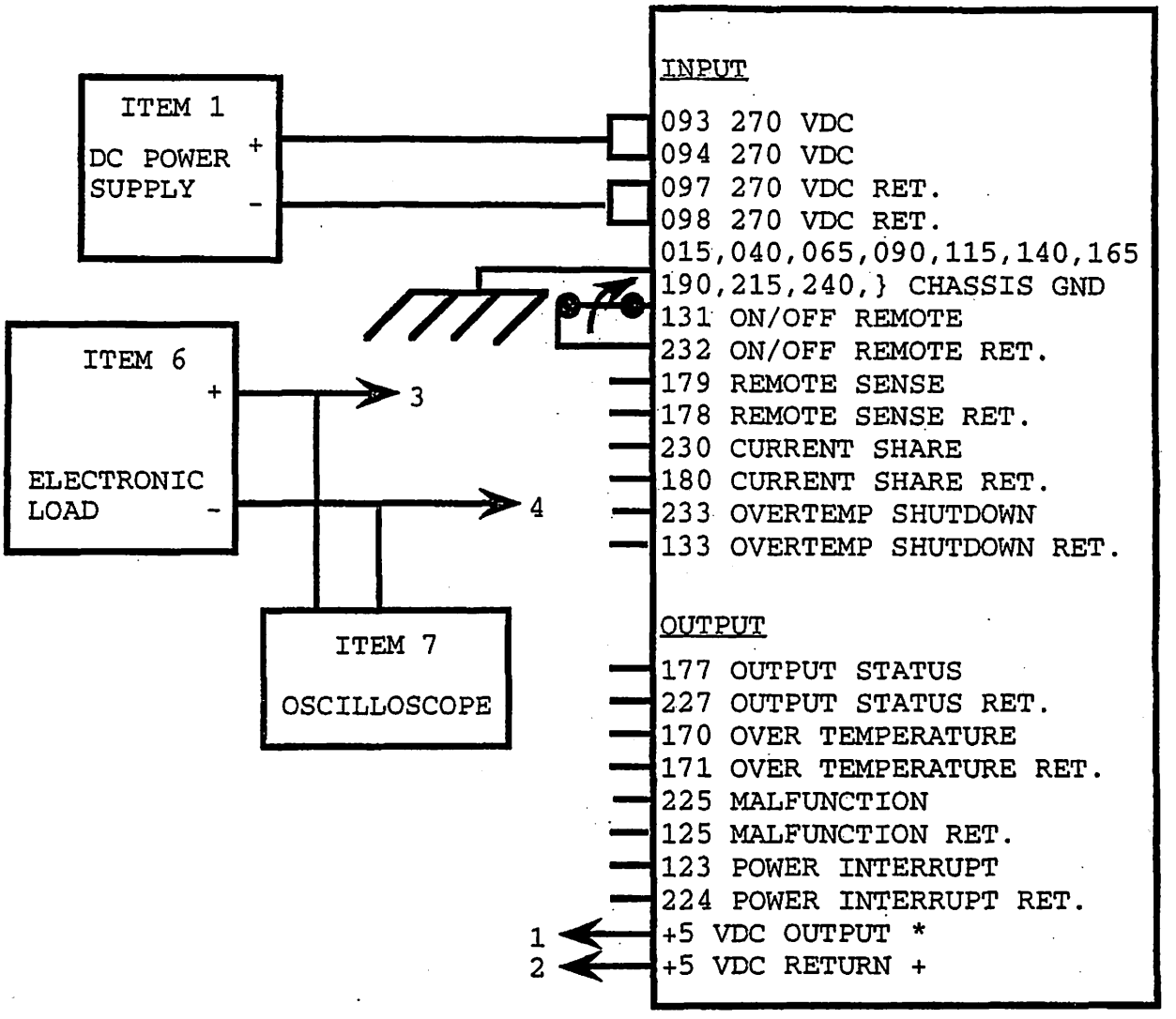
| Test cond | Parameter | Connect | | Item 1 +0/-2 percent | Item 6 +0/-2 percent | Item 4 | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | | Min | Max | | | |
| P4 | Control turn-on transient | 3 | 4 | 270V | 40A | 5V | VOUT | -0.25 | 0.25 | V | 19 | Q |
| P5 | Control turn-on time | 3 | 4 | 270V | 40A | 5V | TON | -- | 250 | msec | 19 | S |
| P6 | Control turn-off transient | 3 | 4 | 270V | 40A | 5V | VOUT | -0.25 | 0.25 | V | 19 | Q |

NOTE:

1. Pin 131 shorted to pin 232 shall enable the unit under test (UUT). Pin 131 and pin 232 shall be opened; UUT shall turn off within the specified limits. Pin 131 shall be switched (shorted) to pin 232; UUT shall turn on with transient within specified limits.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 19. Control turn-on transient, time, and turn-off transient test circuit for table XVII.



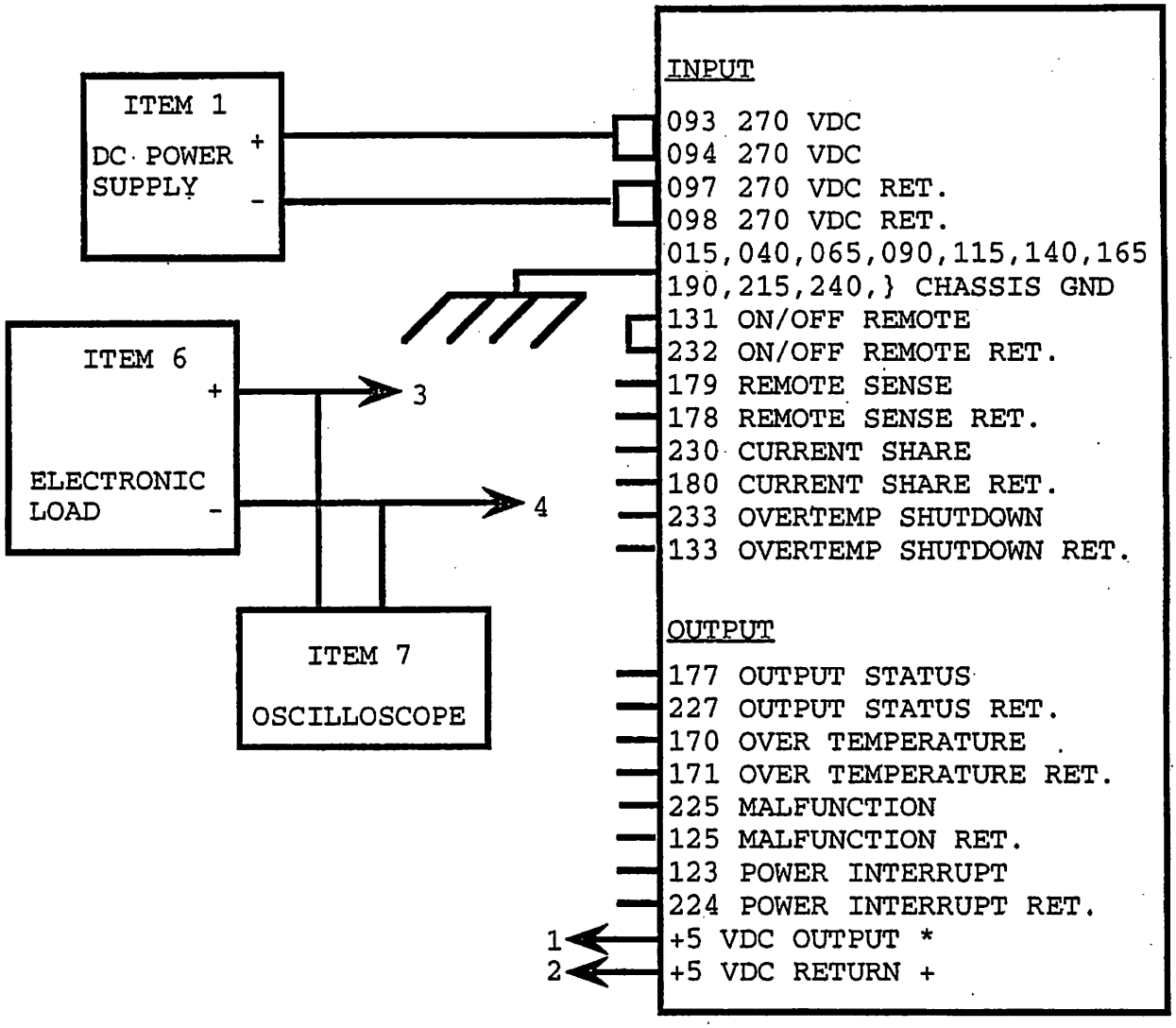
Source: https://assist.dla.mil -- Downloaded: 202640-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Ripple, total, specified load.

| Test cond | Parameter | Connect | | Item 1 +0/ -2 percent | Item 6 +0/ -2 percent | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | Min | Max | | | |
| Q1 | Ripple, total | 3 | 4 | 240V | 4A | Ripple VOUT | -- | 130 | mVp-p | 20 | X |
| Q2 | Ripple, total | 3 | 4 | 280V | 4A | Ripple VOUT | -- | 130 | mVp-p | 20 | X |
| Q3 | Ripple, total | 3 | 4 | 240V | 40A | Ripple VOUT | -- | 130 | mVp-p | 20 | S |
| Q4 | Ripple, total | 3 | 4 | 280V | 40A | Ripple VOUT | -- | 130 | mVp-p | 20 | S |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 20. Ripple, total, specified load test circuit for table XVIII.



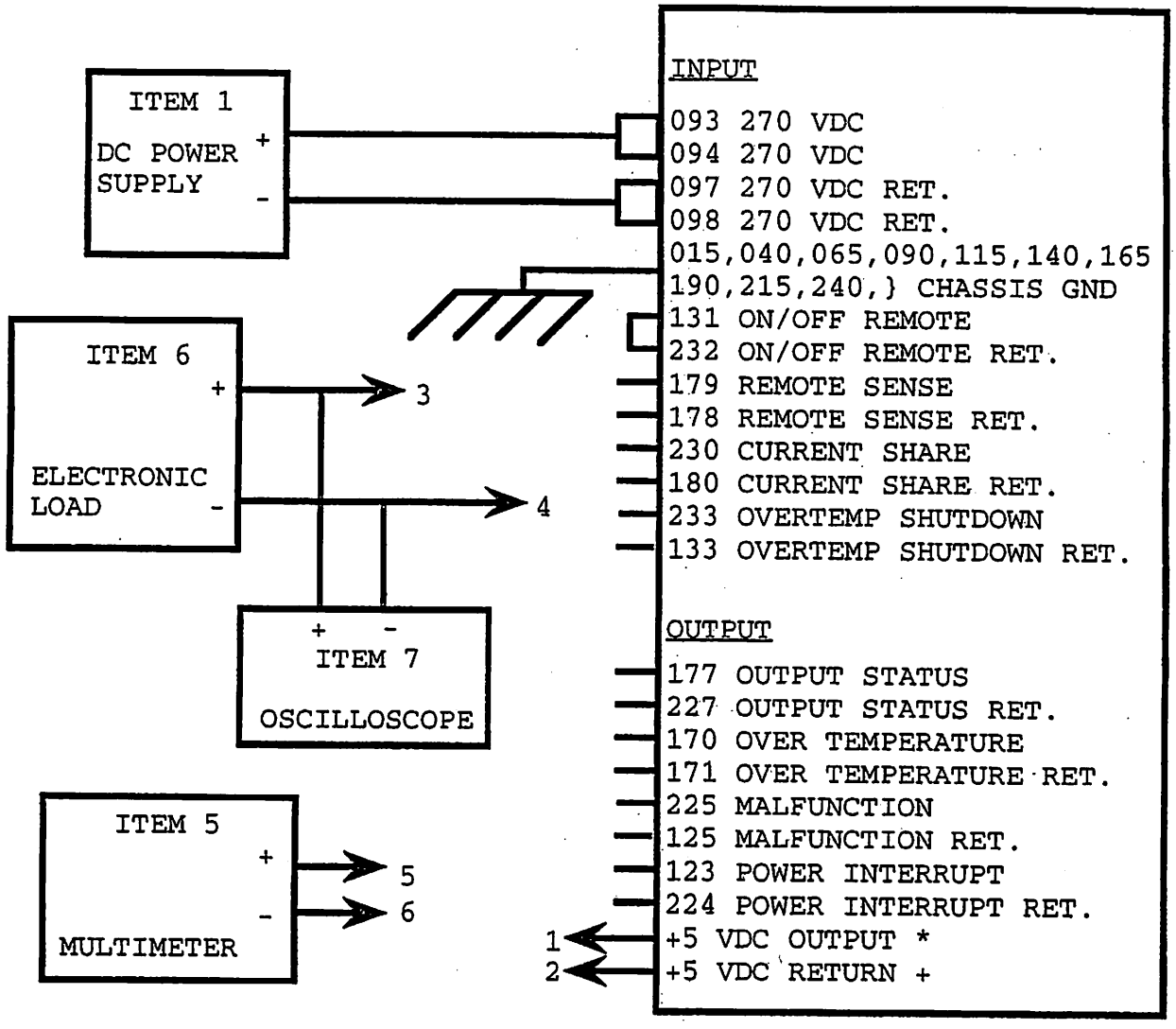
Source: https://assist.dla.mil -- Downloaded: 202640-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XIX. Ripple, total, over and under current.

| Test cond | Parameter | Connect | | Item 1 +0/-2 percent | Item 6 +0/-2 percent | Item 5 | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | | Min | Max | | | |
| Q5 | Ripple, over-current | 3, 5 | 4, 6 | 270V | >40A | ≥4.73V | Ripple VOUT | -- | 200 | mVp-p | 21 | Q |
| Q6 | Ripple, under-current | 3, 5 | 4, 6 | 270V | <4A | -- | Ripple VOUT | -- | 200 | mVp-p | 21 | Q |



Source: https://assist.dla.mil -- Downloaded: 202610-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 21. Ripple, total, over and under current test circuit for table XIX.



Source: https://assist.dla.mil -- Downloaded: 202640-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XX. Total ripple, 10Hz to 20MHz.

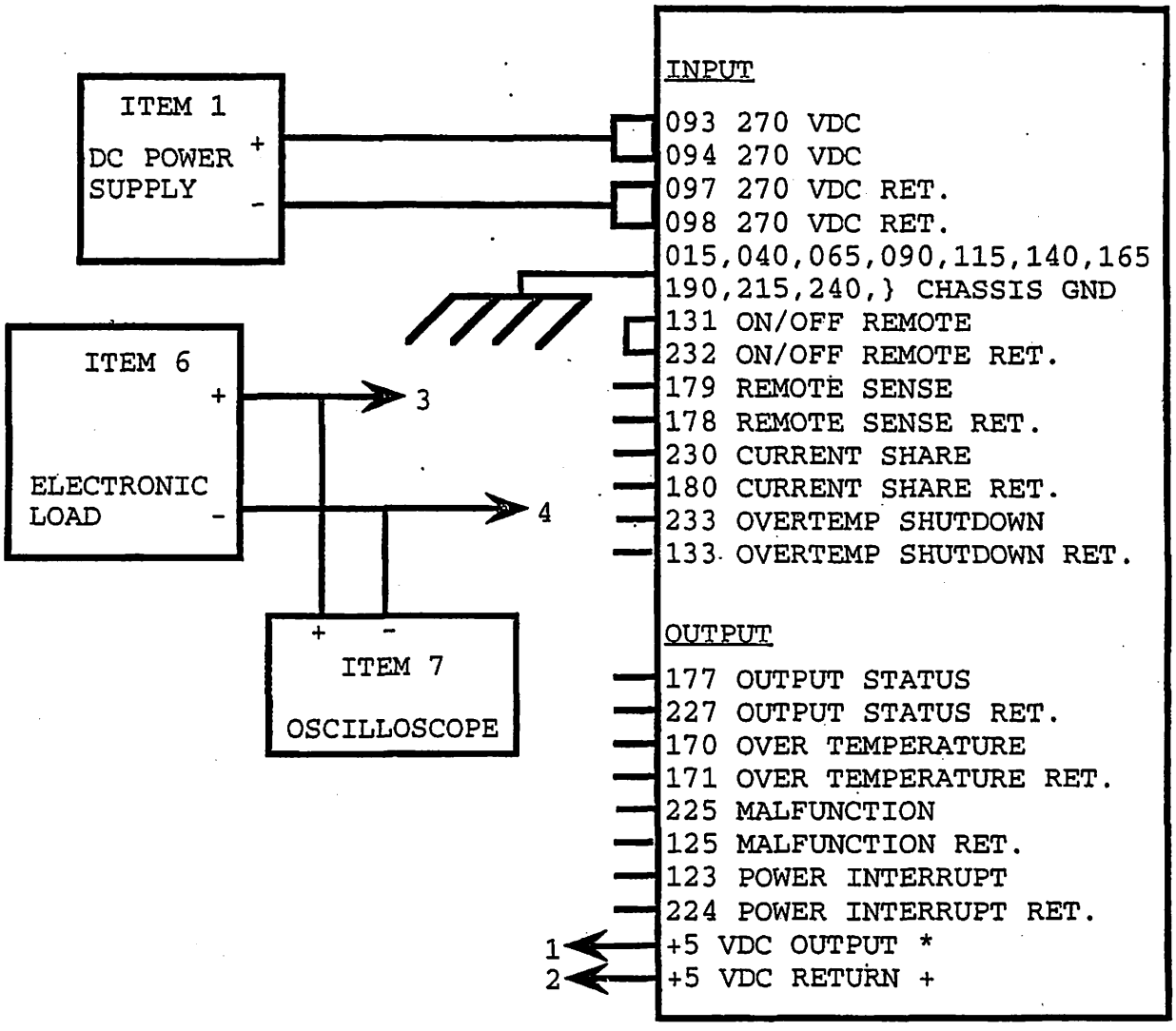
| Test cond | Parameter | Connect | | Item 1 +0/ -2 percent | Item 6 +0/ -2 percent | Item 7 | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | | Min | Max | | | |
| Q7 | Ripple | 3 | 4 | 240V | 40A | 10Hz-20MHz | Ripple VOUT | -- | 300 | mVp-p | 22 | Q |
| Q8 | Ripple | 3 | 4 | 280V | 40A | 10Hz-20MHz | Ripple VOUT | -- | 300 | mVp-p | 22 | Q |

NOTE:

1. Set oscilloscope (item 7) bandwidth at 5Hz to 20MHz.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 22. Total ripple, 10Hz to 20MHz test circuit for table XX.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XXI. Dynamic line regulation, +21 percent.

| Test cond | Parameter | Item 1 +0/ -2 percent | Item 6 +0/ -2 percent | Item 3 | Item 2 | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Min | Max | | | |
| R1 | Dynamic line regulation | 270V | 40A | 4Hz square wave 0V offset, 8V | 56.7V | 1st 1/2 cycle | -- | 150 | mV | 23 | Q |
| R2 | Dynamic line ringing | 270V | 40A | 4Hz square wave 0V offset, 8V | 56.7V | 2nd 1/2 cycle | -- | 75 | mV | 23 | Q |

NOTE:

1. The initial overshoot may be positive or negative.



Source: https://assist.dla.mil -- Downloaded: 202610-21T00:22Z
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TABLE XXII. Dynamic line regulation, -21 percent.

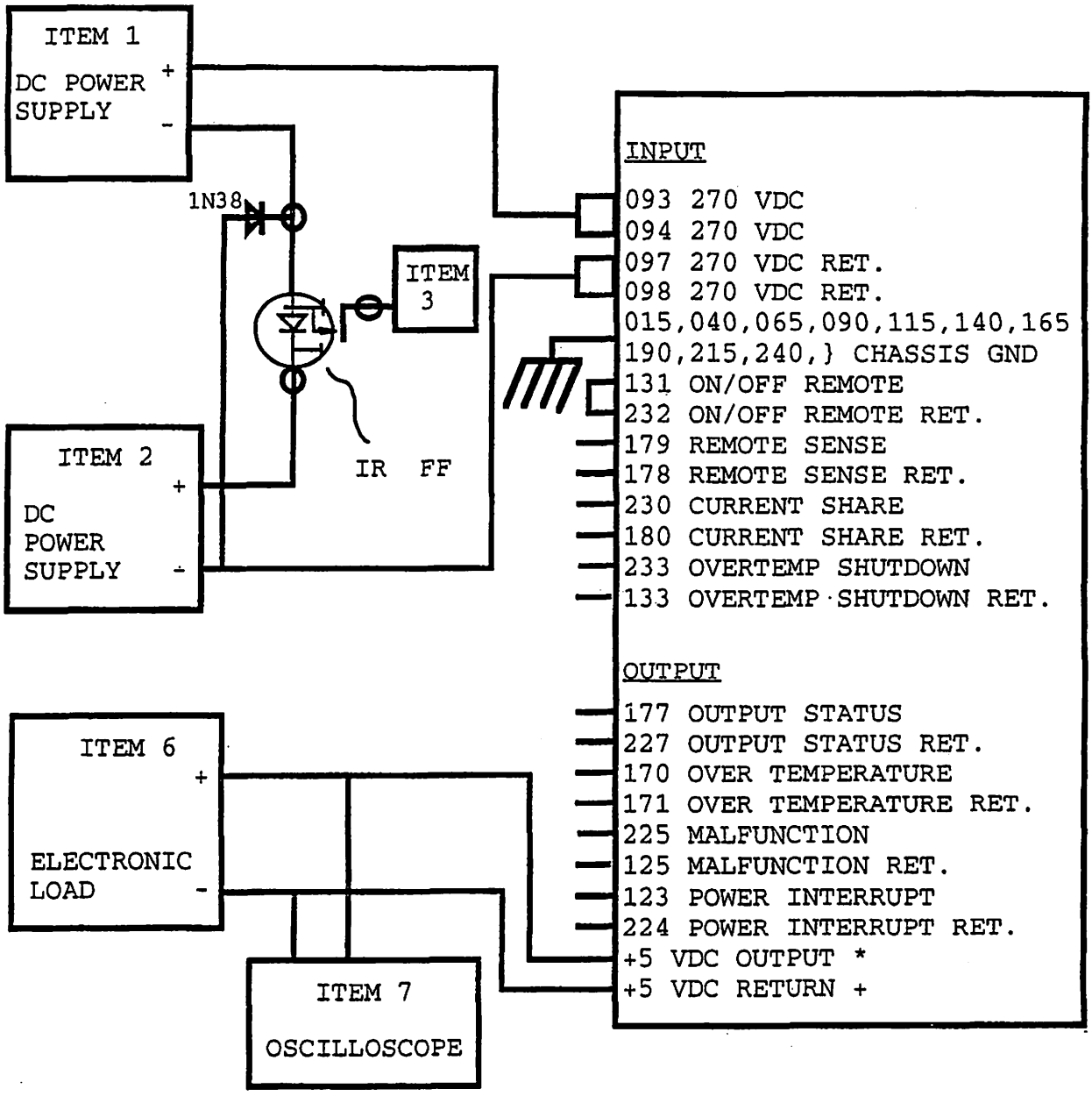
| Test cond | Parameter | Item 1 +0/-2 per-cent | Item 6 +0/ -2 per-cent | Item 3 | Item 2 +0/ -2 per-cent | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Min | Max | | | |
| R3 | Dynamic line regula-tion | 213.3V | 40A | 4Hz sine wave 0V offset, 8V | 56.7V | 1st 1/2 cycle | -- | 150 | mV | 23 | Q |
| R4 | Dynamic line ringing | 213.3V | 40A | 4Hz sine wave 0V offset, 8V | 56.7V | 2nd 1/2 cycle | -- | 75 | mV | 23 | Q |

NOTE:

1. The initial overshoot may be positive or negative.



Source: https://assist.dla.mil -- Downloaded: 202610-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 23. Dynamic line regulation test circuit for tables XXI and XXII.



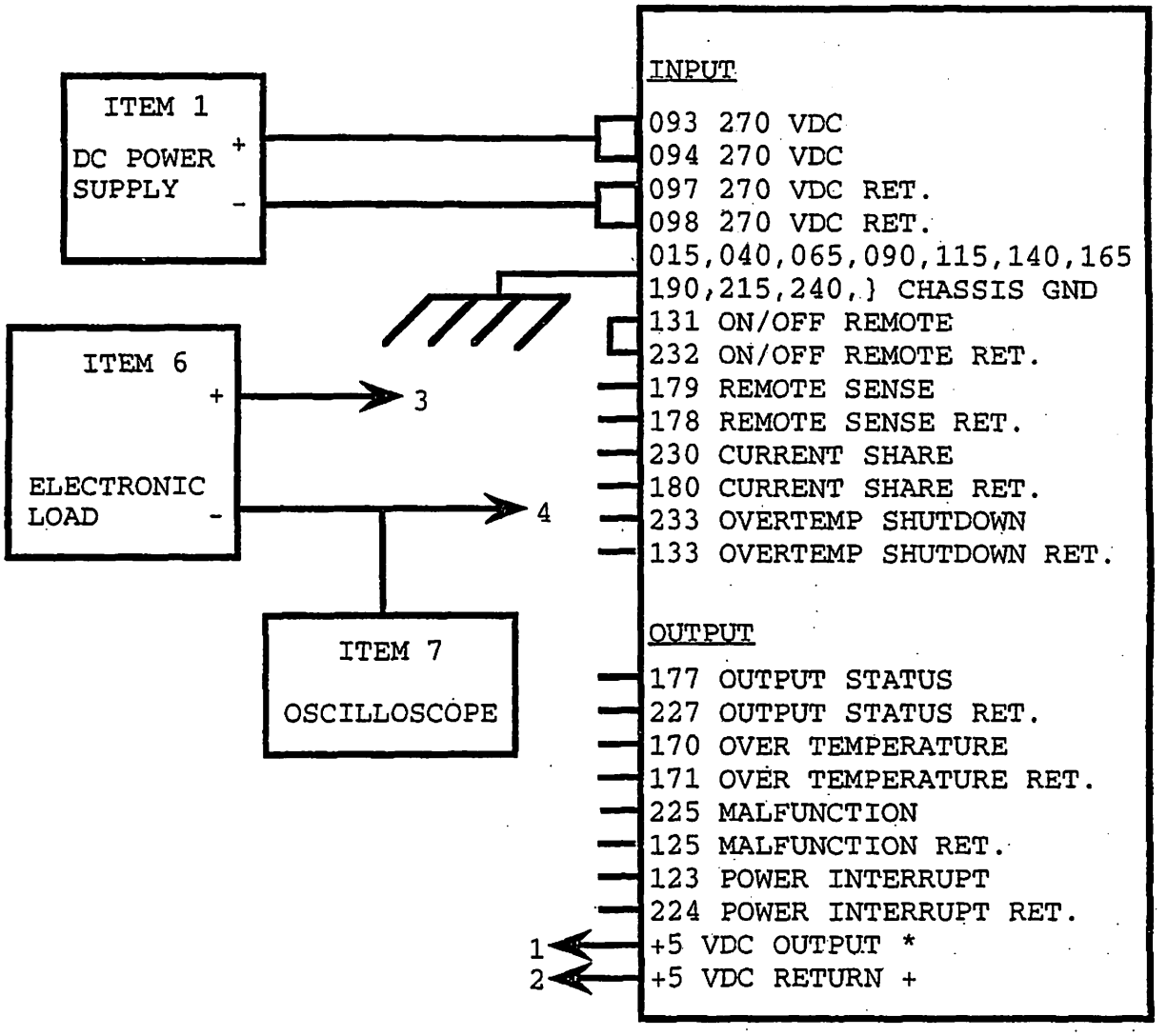
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Dynamic load regulation.

| Test cond | Parameter | Connect | | Item 1 +0/ -2 per-cent | Item 6 | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | Min | Max | | | |
| S1 | Load regulation 80 - 90 percent | 3 | 4 | 270V | 32A to 36A | 1st 1/2 cycle | -- | 150 | mV | 24 | X |
| S2 | Load ringing 80 - 90 percent | 3 | 4 | 270V | 32A to 36A | 2nd 1/2 cycle | -- | 75 | mV | 24 | S |
| S3 | Load regulation 90 - 80 percent | 3 | 4 | 270V | 36A to 32A | 1st 1/2 cycle | -- | 150 | mV | 24 | X |
| S4 | Load ringing 90 - 80 percent | 3 | 4 | 270V | 36A to 32A | 2nd 1/2 cycle | -- | 75 | mV | 24 | S |
| S5 | Load regulation 70 - 90 percent | 3 | 4 | 270V | 28A to 36A | 1st 1/2 cycle | -- | 150 | mV | 24 | Q |
| S6 | Load ringing 70 - 90 percent | 3 | 4 | 270V | 28A to 36A | 2nd 1/2 cycle | -- | 75 | mV | 24 | Q |
| S7 | Load regulation 90 - 70 percent | 3 | 4 | 270V | 36A to 28A | 1st 1/2 cycle | -- | 150 | mV | 24 | Q |
| S8 | Load ringing 90 - 70 percent | 3 | 4 | 270V | 36A to 28A | 2nd 1/2 cycle | -- | 75 | mV | 24 | Q |



Source: https://assist.dla.mil -- Downloaded: 202640-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 24. Dynamic load regulation test circuit for table XXIII.



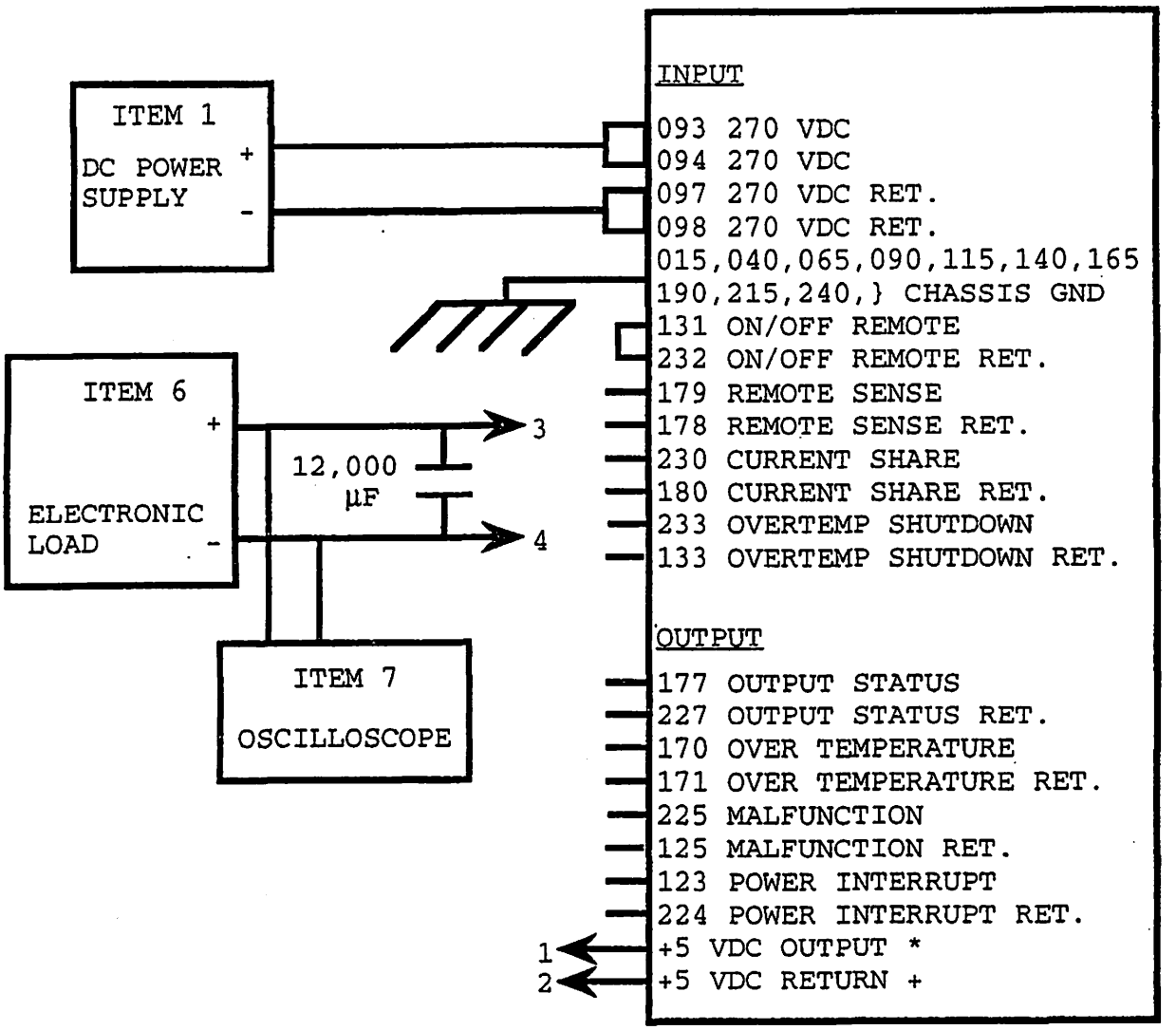
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Dynamic load regulation, 30-80 percent full capacitor load.

| Test cond | Parameter | Connect | | Item 1 +0/ -2 percent | Item 6 | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | Min | Max | | | |
| S9 | Dynamic load regulation, capacitive load | 3 | 4 | 270V | 12A to 32A | VOUT | -- | 150 | mV | 25 | Q |
| S10 | Dynamic load ringing, capacitive load | 3 | 4 | 270V | 32A to 12A | VOUT | 0 | 75 | mV | 25 | Q |



Source: https://assist.dla.mil -- Downloaded: 202640-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 25. Dynamic load regulation, 30-80 percent full capacitor load circuit for table XXIV.



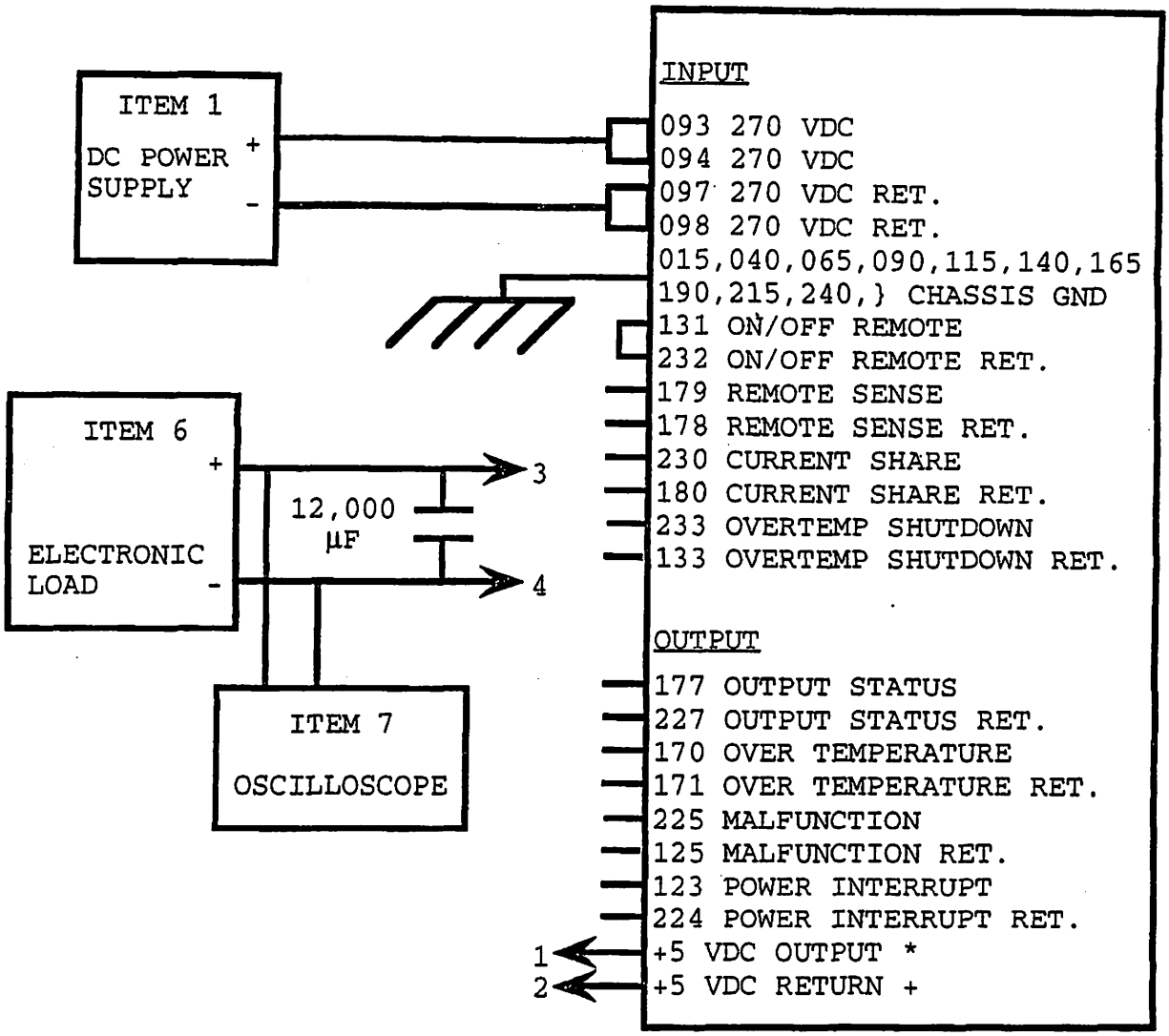
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XXV. Turn-on time into capacitive load.

| Test cond | Parameter | Connect | | Item 1 +0/ -2 percent | Item 6 +0/ -2 percent | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | Min | Max | | | |
| T | Turn on time | 3 | 4 | 270V | 40A | TON | -- | 250 | msec | 26 | Q |



Source: https://assist.dla.mil -- Downloaded: 202610-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 26. Turn-on time into capacitive load circuit for table XXV.



Source: https://assist.dla.mil -- Downloaded: 202610-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Output status signal, logic low.

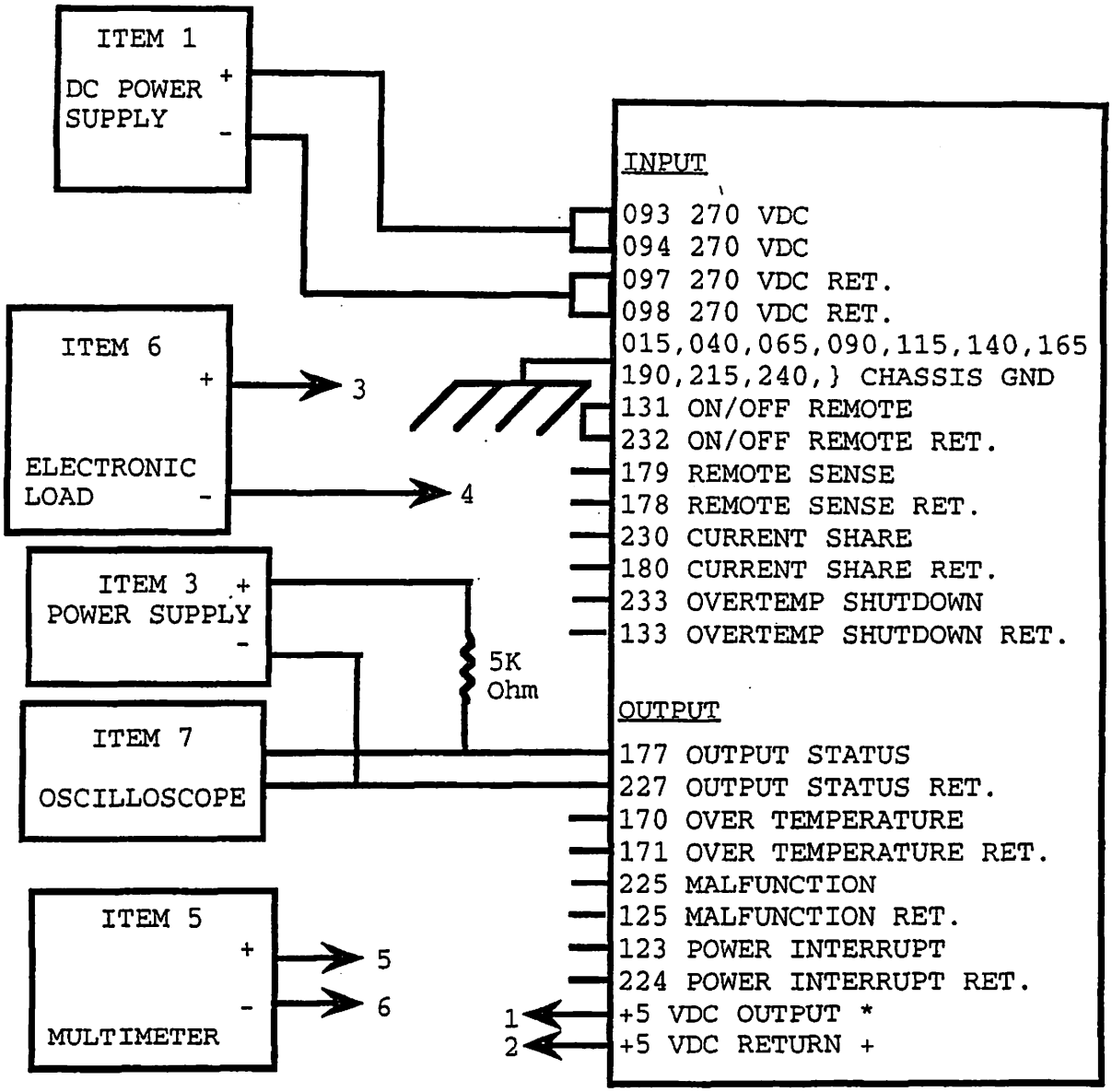
| Test cond | Parameter | Connect | | Item 1 +0/-2 percent | Item 6 +0/-2 percent | Item 5 | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | | Min | Max | | | |
| V1 | Output status signal, 1A | 3,5 | 4,6 | 270V | 1A | 4.64V to 5.56V | Output status | 0.0 | 0.4 | VDC | 27 | X |
| V2 | Output status signal, 40A | 3,5 | 4,6 | 270V | 40A | 4.64V to 5.56V | Output status | 0.0 | 0.4 | VDC | 27 | X |

NOTE:

1. Item 3 on figure 27 shall be set to 30 VDC +0/-2 percent.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 27. Output status test circuit for table XXVI.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
Check the source to verify that this is the current version before use.

TABLE XXVII. Output status signal, logic high.

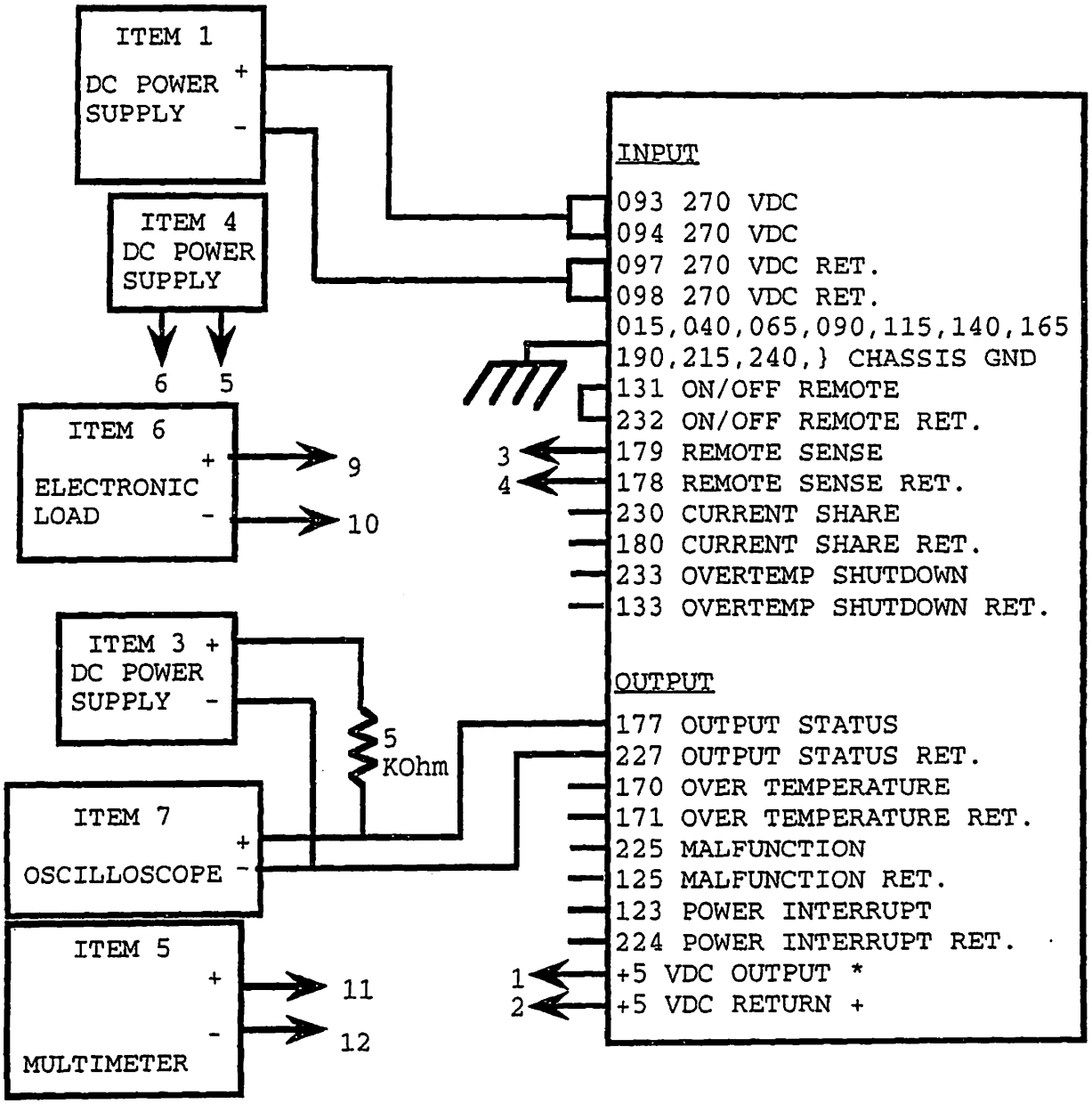
| Test cond | Parameter | Item 1 +0/-2 percent | Item 5 | Item 4 | Item 6 | Connect | | Connect | | Meas | Initial limits | | Unit | Test fig | test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 1 to | 2 to | 3 to | 4 to | | Min | Max | | | |
| V3 | Under voltage | 270V | ≤4.5V | 5.6V | 1A | 9, 11 | 10, 12 | 5 | 6 | Output status | 29.0 | 30.5 | VDC | 28 | X |

NOTE:

1. Item 3 on figure 28 shall be set to 30 VDC +0/-2 percent.



Source: https://assist.dla.mil -- Downloaded: 202610-21T00:22Z
Check the source to verify that this is the current version before use.




* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 28. Output status signal, logic high test circuit for table XXVII.



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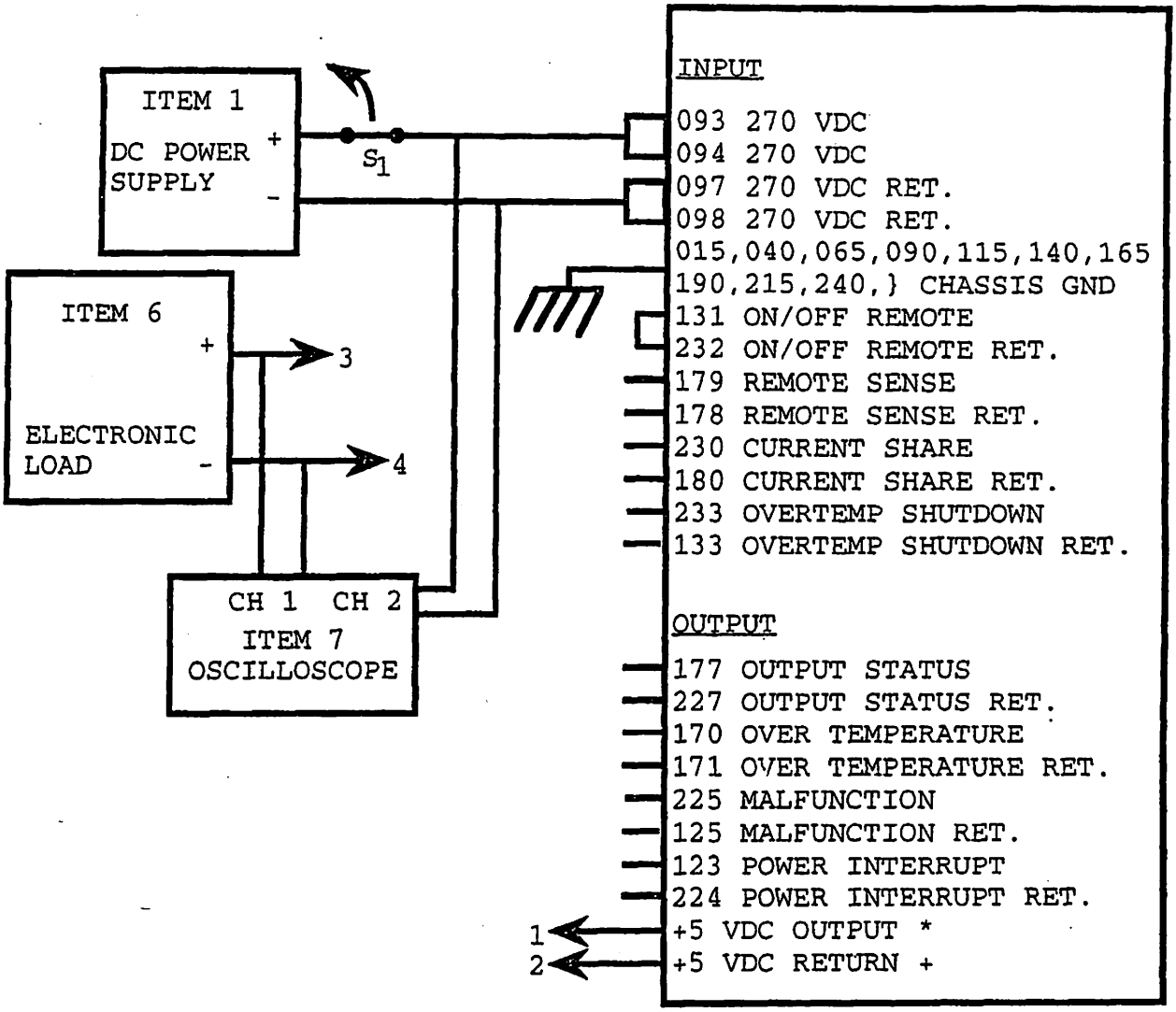
TABLE XXVIII. Output voltage hold-up time.

| Test cond | Parameter | Item 1 +0/-2 percent | Item 6 +0/-2 percent | Connect | | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 1 to | 2 to | | Min | Max | | | |
| X | Output voltage hold-up time | 175V | 40A | 3 | 4 | t hold | 1/ 50 | -- | µsec | 29 | Q |

1/ Hold-up time will be measured from the time power is disconnected from the UUT (t off) to the time VOUT falls to 4.85V (t drop). t hold = t drop - t off.



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* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 29. Output voltage hold-up time test circuit for table XXVIII.



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TABLE XXIX. Input to chassis capacitance.

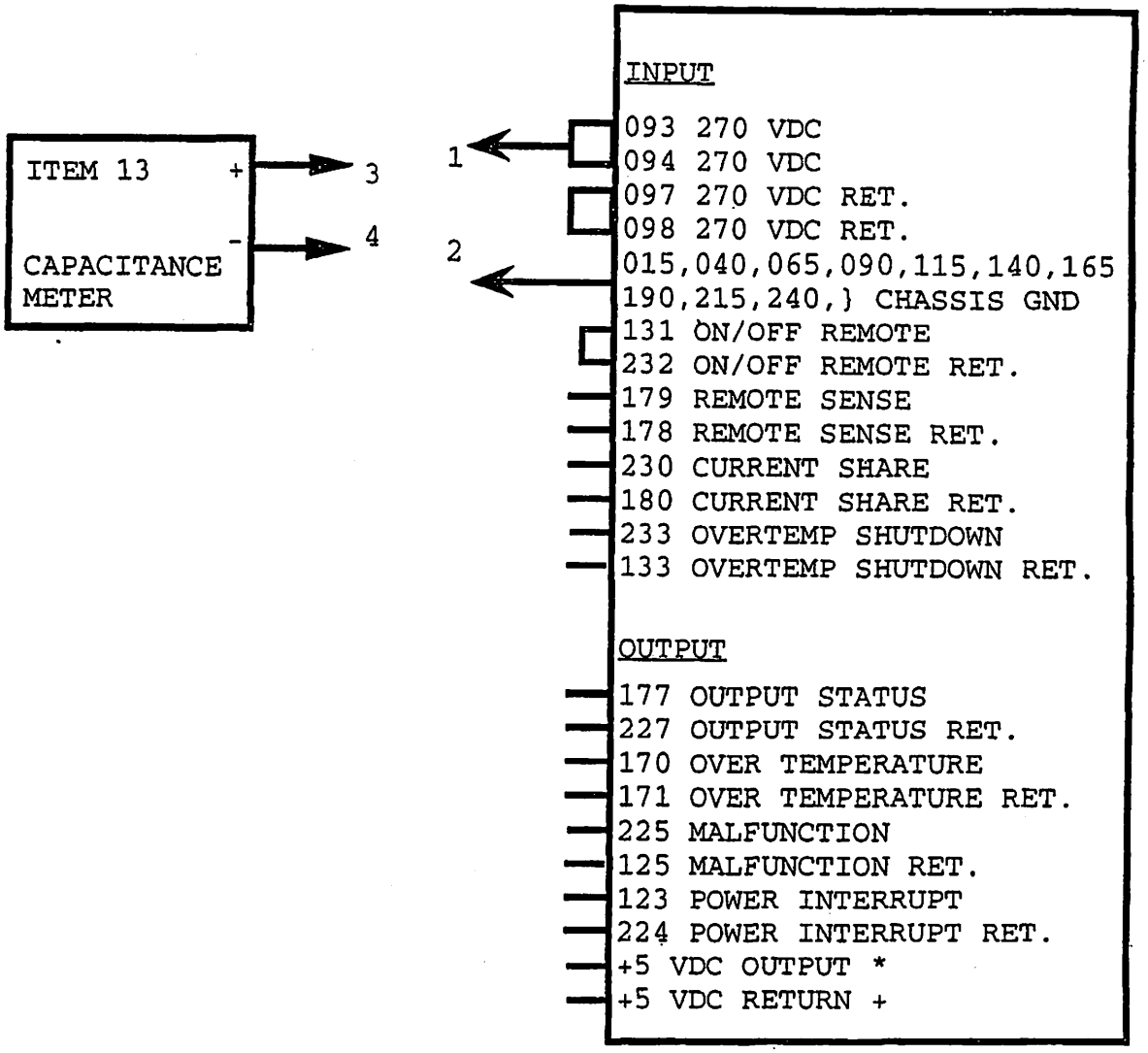
| Test cond | Parameter | Connect | | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | Min | Max | | | |
| Y | Input to chassis capacitance | 3 | 4 | CAP | -- | 0.004 | µF | 30 | S |

NOTE:

1. Capacitance meter shall be set at 1KHz.



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* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 30. Input to chassis capacitance test circuit for table XXIX.



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TABLE XXX. Output impedance, 10Hz - 25KHz, 100 percent load.

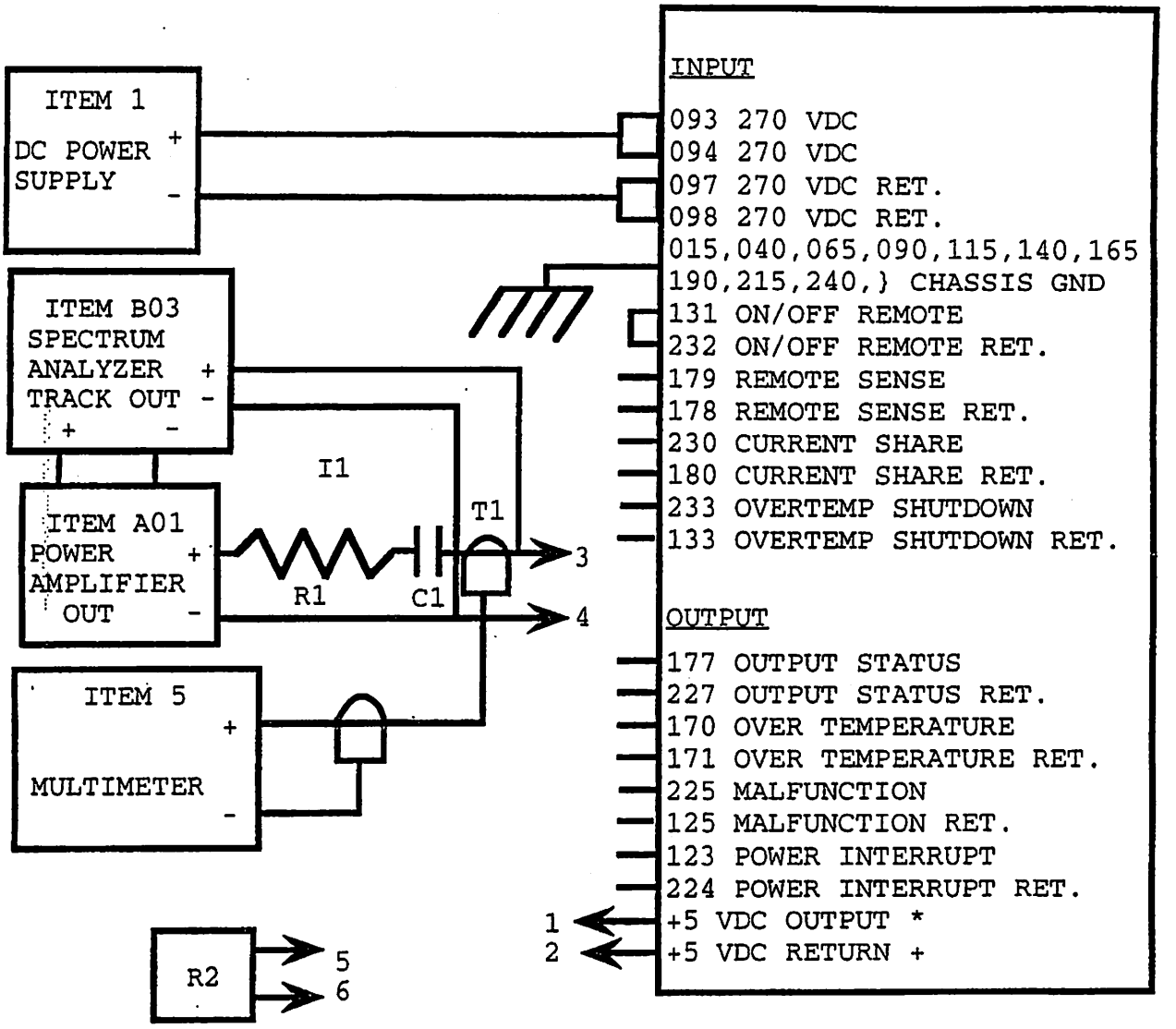
| Test cond | Parameter | Item 2 +0/-2 percent | Item BO3 | Connect | | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 1 to | 2 to | | Min | Max | | | |
| Z | Output impedance, 10Hz to 25 KHz, 100 percent load | 270V | Vo | 3,5 | 4,6 | ZOUT | 0.0 | 30 | mOhm | 31 | Q |

NOTE:

1. See figure 31, note 6.



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* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 31. Output impedance, 10Hz - 25KHz, 100 percent load, test circuit for table XXX.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:22Z
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NOTES:

1. R1 = 10 Ohm ± 5 percent, non-inductive, 0.5 watt resistor.
2. R2 = 0.125 Ohm ± 5 percent, 300 watt resistor, non-inductive.
3. C1 = 6000 µF + 75 percent, - 10 percent, 25 volts capacitor.
4. T1 = Pearson current transformer 411.
5. Monitor shunt voltage with an item 5 voltmeter to ensure that the load current is within specification.
6. The output impedance test shall be performed as follows:

(a) Make the required connections for test condition Z with the item AO1 amplitude turned down to 0V and apply input power to the power supply. The DC load current on the power supply will be 40 amperes ± 10 percent for test condition Z.

(b) AC current adjustment - Set the spectrum analyzer to the manual mode and at a fixed frequency between 100Hz and 1KHz. Use the model 411 current transformer and item 5 to monitor the AC current (conversion factor = 0.1 volts per ampere). Use the item BO3 tracking oscillator level control and the item AO1 amplitude control to increase the AC current into the power supply output to 200 milliampere (mA) ± 3 percent.

(c) Set the item BO3 spectrum analyzer to the sweep mode and find maximum Vo (as measured by item BO3) over the frequency range of 10Hz to 25KHz. Sweep the frequency over a minimum of two bands:

10Hz to 4KHz, resolution bandwidth = 10Hz.
5KHz to 25KHz, resolution bandwidth = 30Hz.

The circuit is set up so the current into the power supply output remains constant over the sweep range. The frequency of maximum Vo is the frequency of maximum output impedance. Read the average value displayed on item BO3. Record the frequency of maximum Vo.

(d) To calculate the maximum output impedance from the maximum Vo-

$$Zo = \frac{\text{item BO3 voltage}}{IIN - \frac{\text{item BO3 voltage}}{R2}} = \frac{\text{item BO3 voltage}}{0.2A - \frac{\text{item BO3 voltage}}{R2}}$$

FIGURE 31. Output impedance, 10Hz - 25KHz, 100 percent load, test circuit for table XXX - Continued.



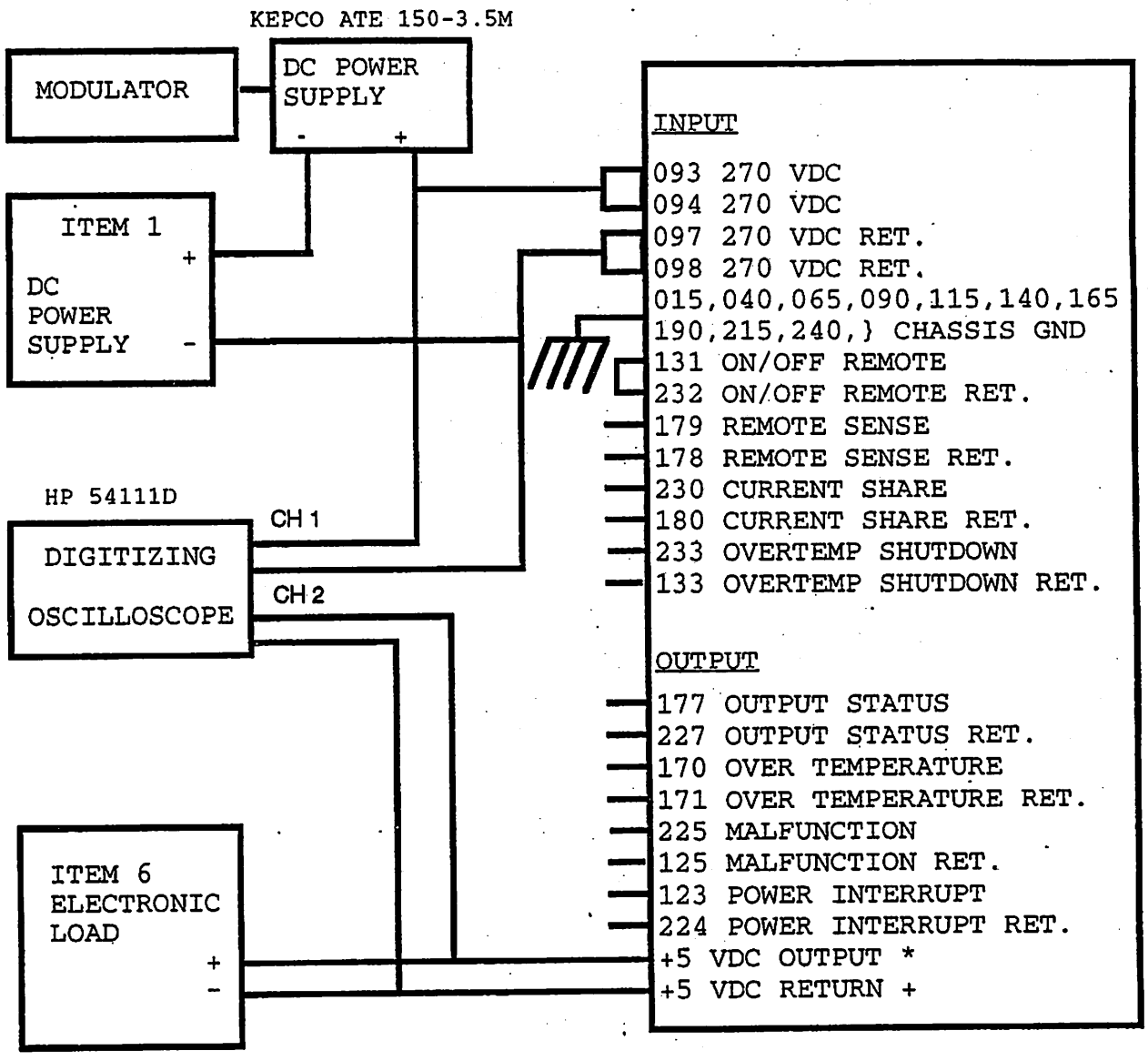
Source: https://assist.dla.mil -- Downloaded: 202610-21T00:22Z
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TABLE XXXI. Line transient response.

| Test cond | Parameter | Item 2 +0/-2 per-cent | Item 3 +0/-2 per-cent | Connect | | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 1 to | 2 to | | Min | Max | | | |
| BB1 | Line transient response (high line) | 280V | 40A | 3 | 4 | Initial tran-sient | -0.25 | 0.25 | V | 32 | Q |
| BB2 | Line transient response (high line), 2nd half cycle | 280V | | 3 | 4 | 2nd half cycle | -0.13 | 0.13 | V | 32 | Q |
| BB3 | Line transient response (low line) | 240V | 40A | 3 | 4 | Initial tran-sient | -0.25 | 0.25 | V | 33 | Q |
| BB4 | Line transient response (low line), 2nd half cycle | 240V | | 3 | 4 | 2nd half cycle | -0.13 | 0.13 | V | 33 | Q |



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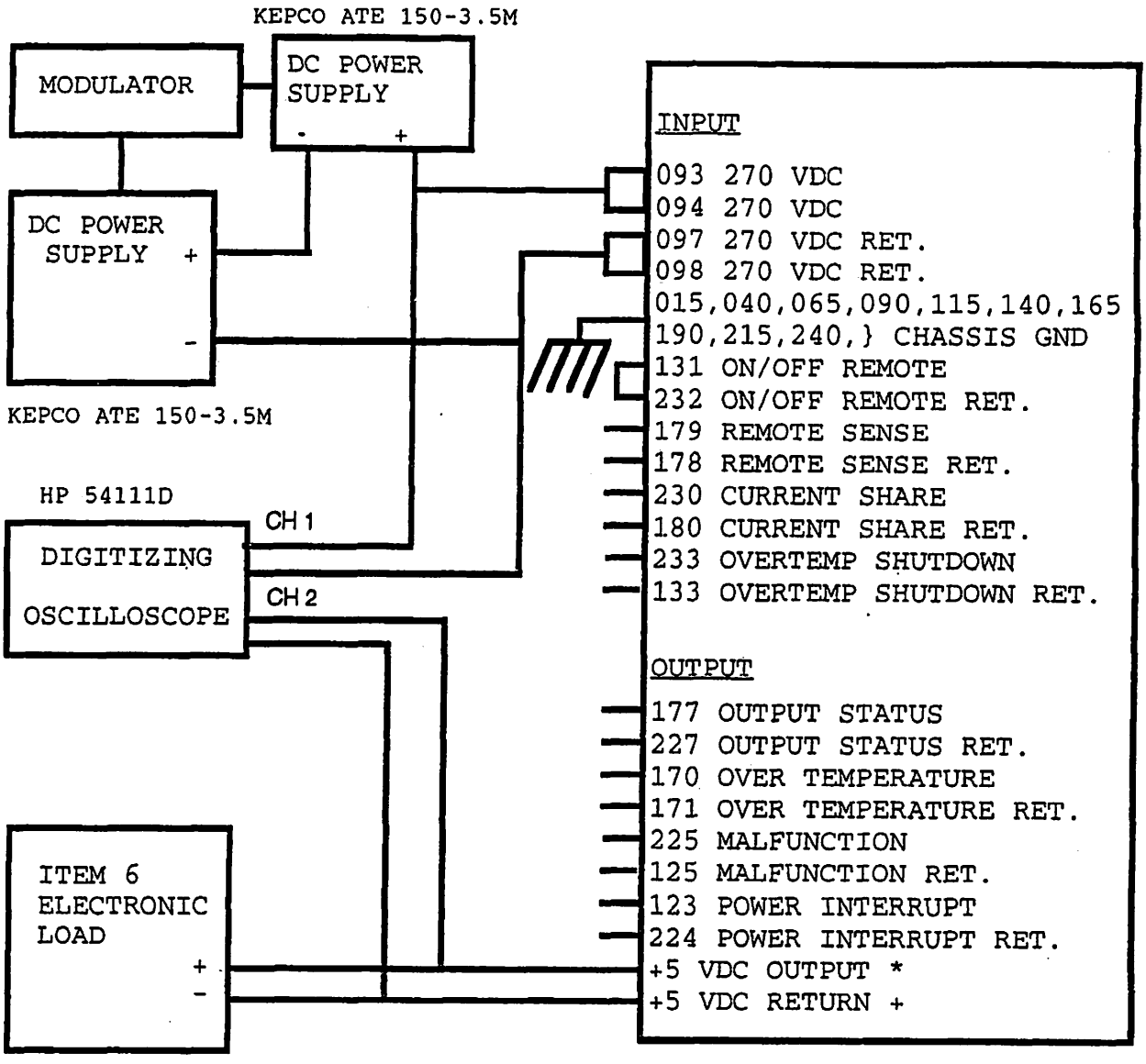

* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 32. Line transient response (high line) test circuit for table XXI.



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* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 33. Line transient response (low line) test circuit for table XXXI.



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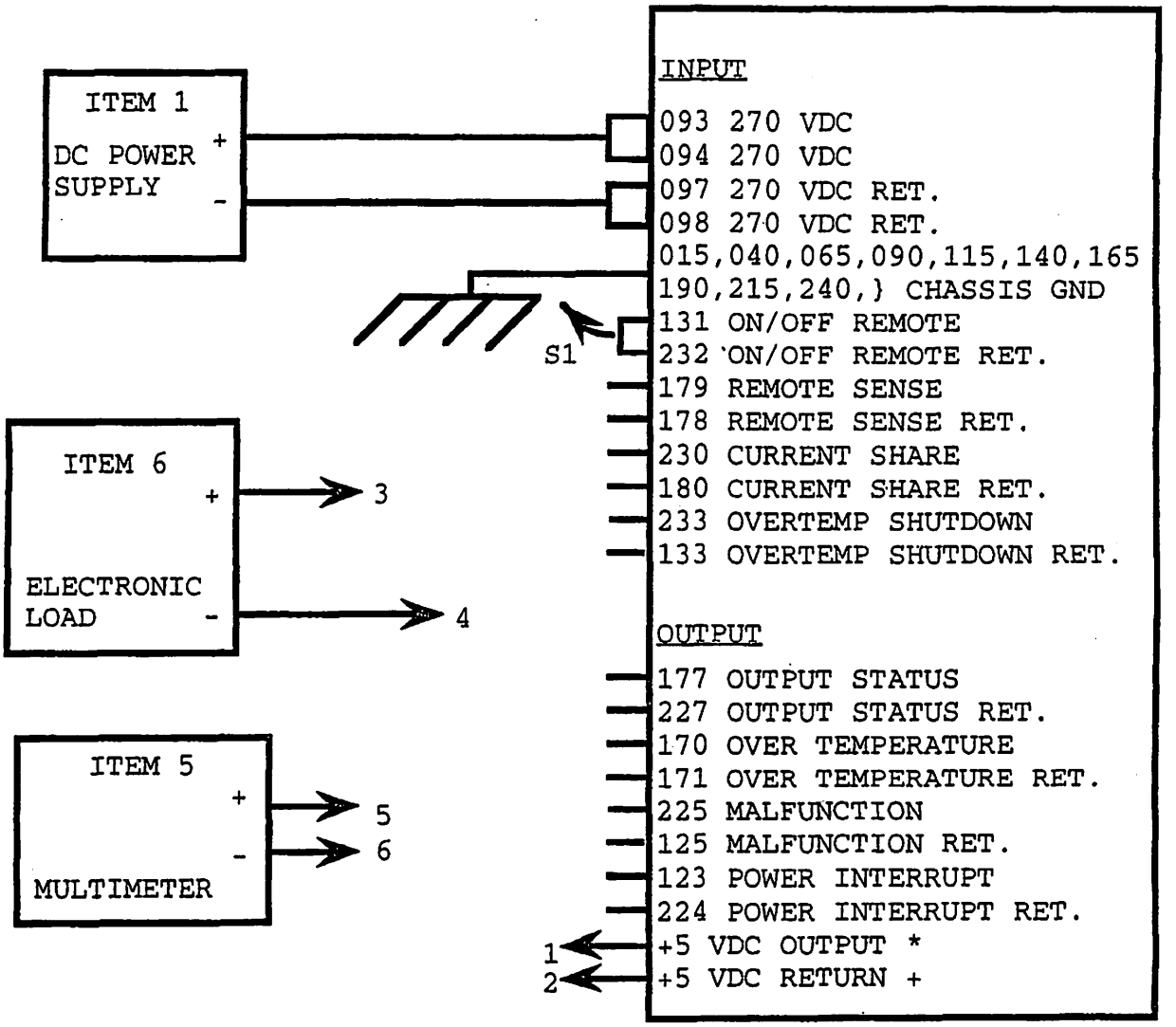
TABLE XXXII. Remote on/off signal open, logic high, logic low.

| Test cond | Parameter | Connect | | Item 1 +0/ -2 percent | Item 6 +0/ -2 percent | Item 5 | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | | Min | Max | | | |
| CC1 | Remote on/off logic low | 3,5 | 4,6 | 270V | 40A | 5V | VOUT | 4.95 | 5.05 | VDC | 34 | Q |
| CC2 | Remote on/off logic high | 3,5 | 4,6 | 270V | 40A | 1/ | VOUT | 0.0 | 0.4 | VDC | 34 | Q |
| CC3 | Remote on/off short | 3,5 | 4,6 | 270V | 40A | 2/ | VOUT | 4.95 | 5.05 | VDC | 34 | Q |

1/ Pins 131 and 232 shall be open for this procedure.
2/ Pins 131 and 232 shall be shorted for this procedure.



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* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 34. Remote on/off signal test circuit for table XXXII.



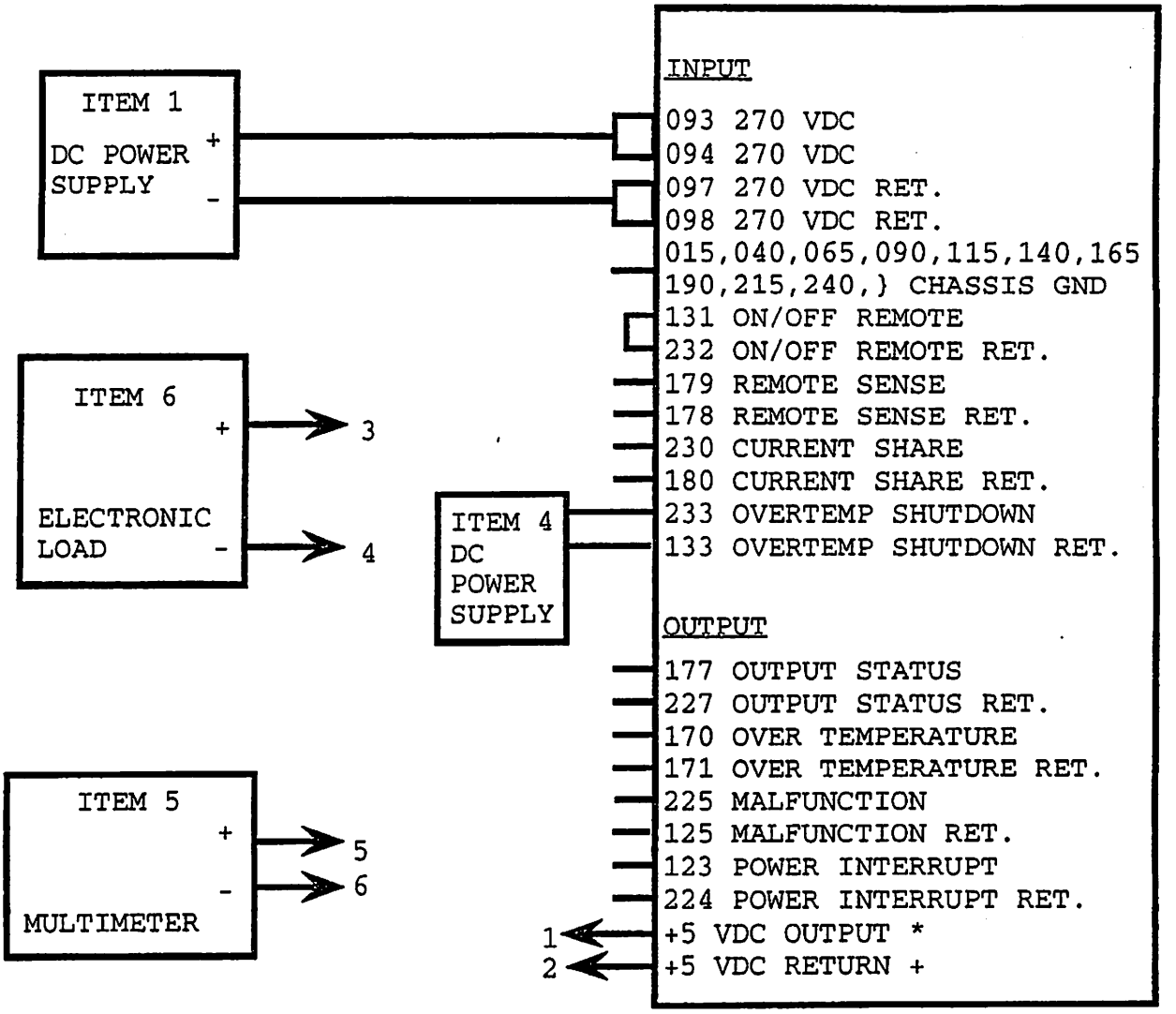
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TABLE XXXIII. Over-temperature shutdown tests.

| Test cond | Parameter | Connect | | Item 1 +0/ -2 percent | Item 6 +0/ -2 percent | Item 4 | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | | Min | Max | | | |
| CC4 | On/off signal, logic high | 3,5 | 4,6 | 270V | 40A | 5.0V | VOUT | 4.95 | 5.05 | VDC | 35 | Q |
| CC5 | On/off signal, logic low | 3,5 | 4,6 | 270V | 40A | 0V | VOUT | 0.0 | 0.4 | VDC | 35 | Q |



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* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 35. Over-temperature shutdown test circuit for table XXXIII.



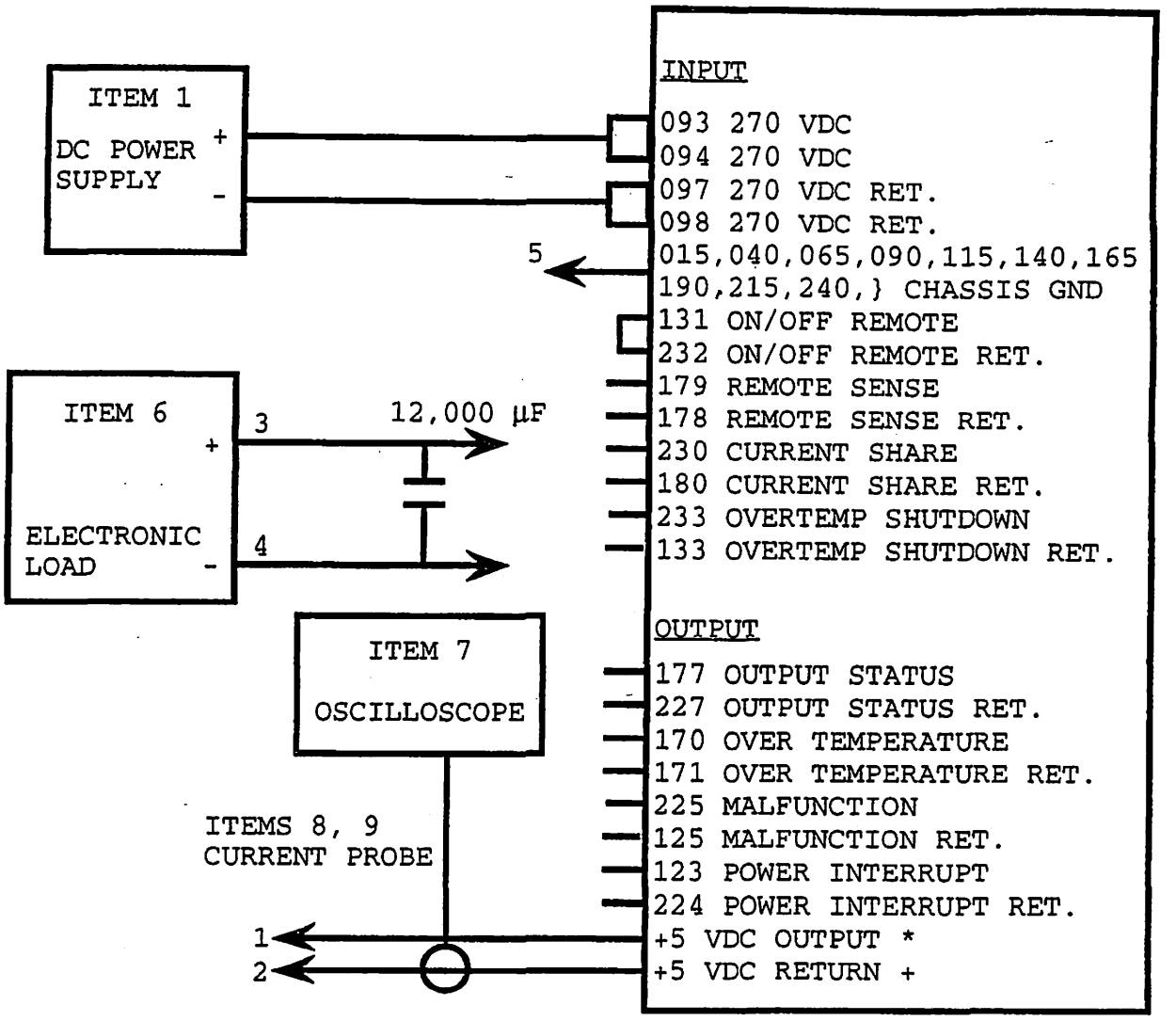
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TABLE XXXIV. Common mode current broadband.

| Test cond | Parameter | Connect | | Item 1 +0/-2 percent | Item 6 +0/-2 percent | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | Min | Max | | | |
| EE1 | Common mode current 40A | 3 | 4,5 | 270V | 40A | ICM | -- | 200 | mAp-p | 36 | Q |
| EE2 | Common mode current 20A | 3 | 4,5 | 270V | 20A | ICM | -- | 200 | mAp-p | 36 | Q |



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* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 36. Common mode current broadband test circuit for table XXXIV.



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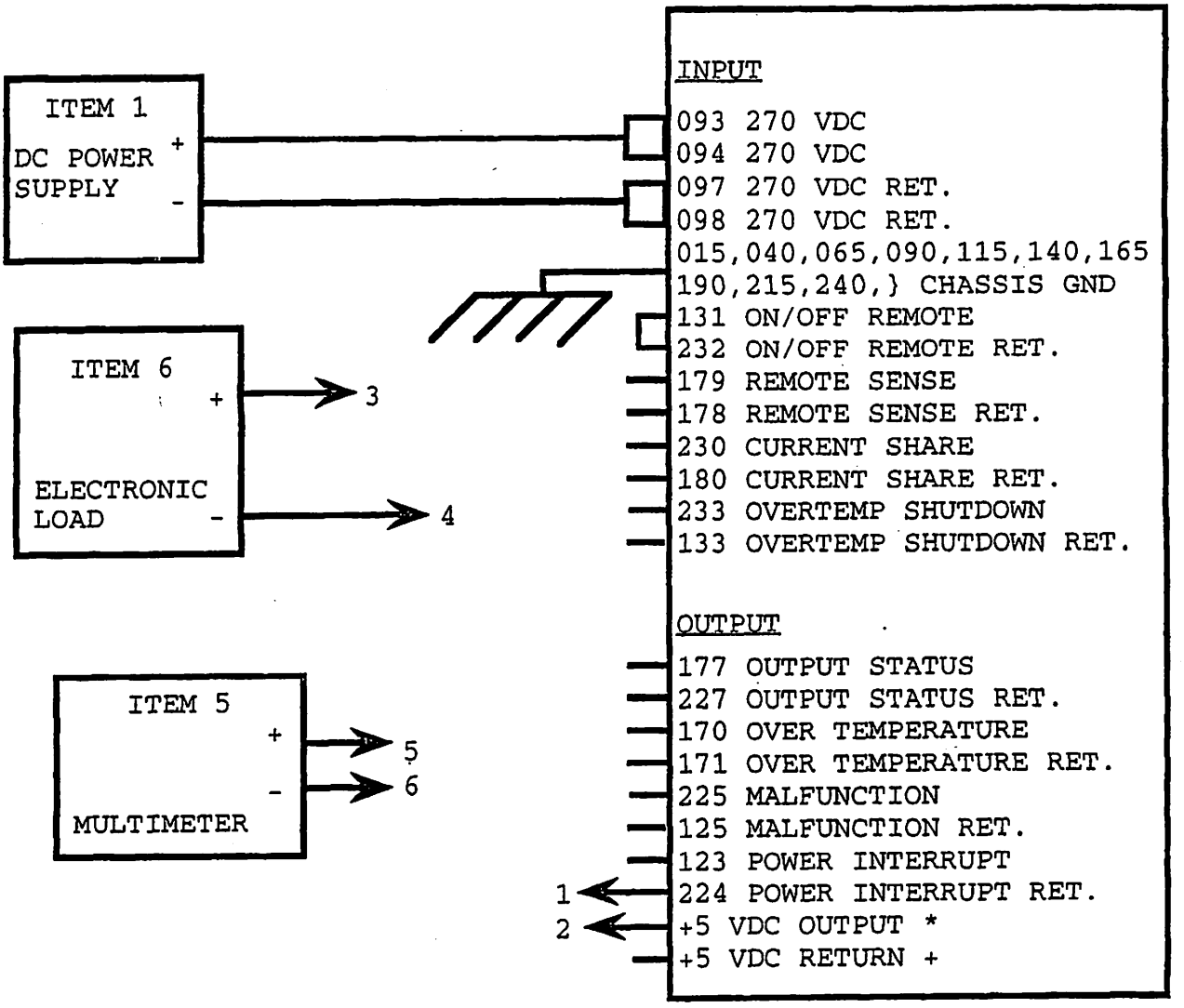
TABLE XXXV. Input voltage turn on/off thresholds.

| Test cond | Parameter | Connect | | Item 1 | Item 6 +0/-2 percent | Item 5 | Meas | Initial limits | | Unit | Test fig | Test type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 to | 2 to | | | | | Min | Max | | | |
| GG1 | Input voltage turn on threshold | 3,5 | 4,6 | 1/ | 40A | 4.75V min | VOUT | 200 | 220 | VDC | 37 | S |
| GG2 | Input voltage turn off threshold | 3,5 | 4,6 | 1/ | 40A | 4.8V | VOUT | -- | 165 | VDC | 37 | S |

1/ Power supply shall be enabled at 200V. Slowly raise supply until VOUT = 4.75V. "V turn-on" is equal to the input voltage. It shall fall between 200V and 220V. Lower the supply until VOUT = 4.8V. "V turn-off" is equal to the input voltage. It shall be 165V or less.



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* +5V CONNECT PINS: 001,002,003,004,005,006,007,008,009,029,030,031, 032,033,034,035,036,037,051,052,053,054,055,056,057,058,059,079,080,081, 082,083,084,085,086,087,102,104,106,108,110,130,132,134,136

\+ +5V RET CONNECT PINS: 010,011,012,013,014,016,017,018,019,020,021, 022,023,024,025,026,027,028,060,061,062,063,064,066,067,068,069,070,071, 072,073,074,075,076,077,078,112,114,116,118,120,122,124,126,128

FIGURE 37. Input voltage turn on/off thresholds test circuit for table XXXV.



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Custodians:
Army - ER
Navy - AS
Air Force - 99

Review activity:
DLA - GS

Preparing activity:
Navy - AS

(Project 6130-0323)

Agent: NW



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